Applied Cryptography: How It Works

Contents

Notation (list of symbols)

k	secret key, shared by encryptor and decryptor (symmetric).
P	plaintext: the message to encrypt, a bit string.
C	ciphertext: the encrypted output.
b	block size in bits (e.g. 128 for AES).

1 Block ciphers vs. stream ciphers

A cryptographic hash is *one-way*: you compute a digest and can never recover the message. *Encryption* is the opposite — reversible, with a secret key — and its goal is *confidentiality*: scramble a message so that only a key-holder can read it.

1.1 Symmetric encryption in one line

Symmetric means a single shared secret key k runs both directions:

$$C = \mathrm{Enc}_k(P), \qquad P = \mathrm{Dec}_k(C).$$

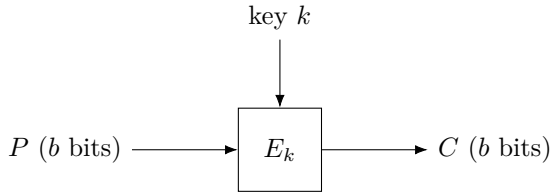
P is the *plaintext* (the message, a bit string), C the *ciphertext* (the scrambled output), and k the key. Dec_k is the exact inverse of Enc_k, so decrypting what you encrypted returns the original. Two families build Enc: *block* ciphers and *stream* ciphers.

1.2 Block ciphers

A block cipher encrypts a *fixed-size block* of bits at a time. Formally it is a keyed *permutation* — a reversible shuffle —

$$E_k : \{0,1\}^b \to \{0,1\}^b,$$

where b is the block size (AES uses $b = 128$ bits). "Permutation" means that for a fixed key, E_k is a bijection on the 2^b possible blocks (one-to-one and onto), so it has an inverse $D_k = E_k^{-1}$ that decryption uses. AES, and the retired DES, are block ciphers.



One block is only b bits, so to encrypt a longer message you need a *mode of operation*: a recipe for chaining many block-cipher calls (CBC, CTR, GCM, ...). Modes usually also require *padding* to fill out the final partial block. (The naive mode, ECB, encrypts each block independently and leaks repeated-block patterns — never use it.)

1.3 Stream ciphers

A stream cipher instead generates a long pseudorandom *keystream* from the key and a *nonce*, and combines it with the plaintext one bit (or byte) at a time with XOR:

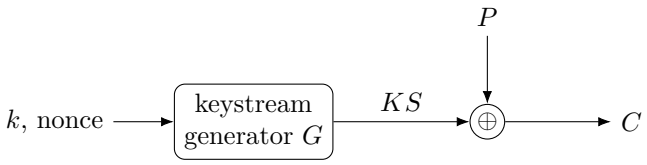
$$KS = G(k, \mathrm{nonce}), \qquad C_i = P_i \oplus KS_i, \qquad P_i = C_i \oplus KS_i.$$

The jargon:

- $\oplus$ (*XOR*, exclusive-or): a bit operation that outputs 1 exactly when its two input bits differ. Its key property is being self-inverse, $x \oplus y \oplus y = x$ — which is why XORing the ciphertext with the *same* keystream gives the plaintext back.
- *keystream* KS: the pseudorandom bit sequence the cipher produces to mask the message.

- *nonce* ("number used once", a.k.a. IV): a non-secret value that makes each encryption under the same key produce a *different* keystream. It must never repeat for a given key (see the pitfall below).

This is a practical *one-time pad*: the one-time pad XORs the message with a *truly random* pad as long as the message — perfectly secret but impractical (the key is as big as the data and usable once); a stream cipher replaces that true-random pad with a pseudorandom keystream stretched from a short key. ChaCha20 is the modern choice; RC4 is the well-known broken one.



1.4 Side by side

	Block cipher	Stream cipher
Unit	fixed b-bit block	bit / byte at a time
Mechanism	keyed permutation E_k	XOR with a keystream
Long messages	need a mode (CBC, CTR, GCM)	naturally continuous
Padding	usually required	none
Examples	AES, DES (retired)	ChaCha20, RC4 (broken)
Random access	CTR yes, CBC no	yes

The line blurs. Run a block cipher in *counter (CTR)* mode — encrypt a running counter to make a keystream $KS_i = E_k(\text{nonce} \,\|\, i)$ and XOR it into the plaintext — and the block cipher *becomes* a stream cipher. So "block vs. stream" is partly about the underlying primitive and partly about how you use it.

The one fatal pitfall: never reuse a (key, nonce) pair. If two messages are XORed with the *same* keystream, then

$$C \oplus C' = (P \oplus KS) \oplus (P' \oplus KS) = P \oplus P',$$

the keystream cancels and the attacker learns $P \oplus P'$ — usually enough to unravel both messages. (This is the "two-time pad" break.) A fresh nonce per message avoids it.

1.5 Symbols

New persistent symbols this section — k, P, C, b — are in the global list of symbols. The symbols below are *local to this section.*

Local symbol	Meaning (this section)
$\mathrm{Enc}_k, \mathrm{Dec}_k$	encrypt / decrypt under key k (exact inverses)
E_k, D_k	block-cipher permutation and its inverse, on b-bit blocks
G	keystream generator
KS	keystream (pseudorandom bits); KS_i is its i-th unit
nonce	"number used once": non-secret, must not repeat per key
$\oplus$	bitwise XOR
P_i, C_i	i-th bit/byte of plaintext / ciphertext
i	position index

2 Modes of operation

A block cipher E_k encrypts exactly one b-bit block (§1). Real messages are longer, shorter, or not a clean multiple of b. A *mode of operation* is the recipe for calling E_k repeatedly to encrypt a message of any length. Throughout, pad the message (§2.4) and split it into n blocks

$$P_1, P_2, \ldots, P_n, \qquad \text{each } b \text{ bits},$$

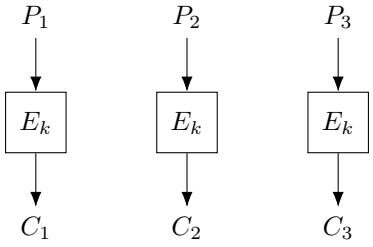
encrypting to ciphertext blocks $C_1, \ldots, C_n$. The modes differ in *how* each block is fed through E_k — and that choice decides everything about safety and speed.

2.1 ECB — and why never to use it

The obvious recipe, *Electronic CodeBook* (ECB), encrypts each block on its own:

$$C_i = E_k(P_i), \qquad P_i = D_k(C_i),$$

where $D_k = E_k^{-1}$ is the inverse permutation (§1).



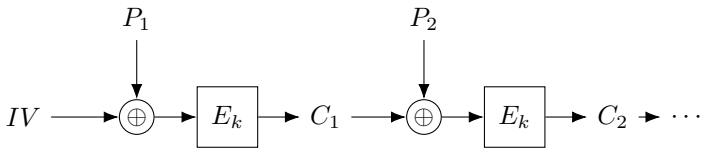
Because E_k is deterministic, *equal plaintext blocks always give equal ciphertext blocks*: if $P_i = P_j$ then $C_i = C_j$. So the ciphertext still shows wherever the message repeats. Encrypt a bitmap image this way and the picture stays recognisable — the infamous "ECB penguin." ECB leaks structure; never use it for real data. The fix in every mode below is to make encryption *randomised* so identical blocks diverge.

2.2 CBC — cipher block chaining

CBC removes the leak by XORing each plaintext block with the *previous ciphertext block* before encrypting, seeding the chain with a random *initialization vector* IV:

$$C_i = E_k\big(P_i \oplus C_{i-1}\big), \qquad C_0 \equiv IV.$$

The IV is a non-secret, random b-bit value chosen fresh per message (like the nonce of §1); for CBC it must be *unpredictable*. Because the IV changes, the same plaintext encrypts to different ciphertext each time — the randomisation ECB lacked.



To decrypt, run D_k and undo the XOR. Starting from $C_i = E_k(P_i \oplus C_{i-1})$, apply D_k to both sides: $D_k(C_i) = P_i \oplus C_{i-1}$; then XOR C_{i-1} and use $x \oplus y \oplus y = x$:

$$P_i = D_k(C_i) \oplus C_{i-1}.$$

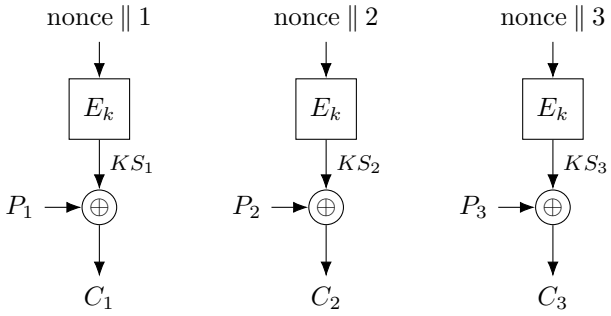
Trade-offs. Encryption is *sequential* — block i needs C_{i-1}, so it cannot be parallelised (decryption can, since all C blocks are already in hand). The message must be padded to a whole number of blocks (below).

2.3 CTR — counter mode

CTR takes the trick from §1: don't encrypt the plaintext at all — encrypt a *counter* to manufacture a keystream, then XOR. For block i,

$$KS_i = E_k(\text{nonce} \,\|\, i), \qquad C_i = P_i \oplus KS_i, \qquad P_i = C_i \oplus KS_i,$$

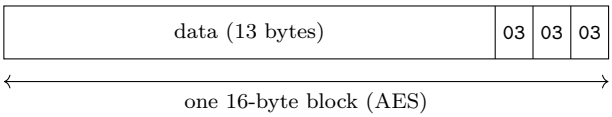
where nonce $\|\, i$ is the nonce concatenated with the block index i, forming a distinct b-bit input for every block, and KS_i is the resulting keystream block.



Trade-offs. The blocks are independent, so CTR is *fully parallel* and allows *random access* (jump straight to block i). It needs *no padding* (truncate the last keystream block to the message length), and uses only E_k — never the inverse D_k. Encryption and decryption are the *same* operation. The one rule, exactly as in §1: never reuse a (key, nonce) pair, or the keystream repeats and the "two-time pad" break applies.

2.4 Padding the last block

ECB and CBC need the plaintext to fill whole b-bit blocks; CTR and stream ciphers do not. The standard scheme, *PKCS#7*: if p bytes are missing from the final block, append p bytes each holding the *value* p. The value records how many to strip, so removal is unambiguous. If the message already fills the blocks exactly, append one whole extra block of padding — so there is always something to remove.



Here 13 data bytes leave $p = 3$ to fill a 16-byte block, so three `0x03` bytes are appended.

2.5 These modes hide, but do not protect

ECB, CBC, and CTR give *confidentiality* — they scramble the message — but no *integrity*: nothing detects tampering. CTR makes this stark. Since $P_i = C_i \oplus KS_i$, flipping any bit of C_i flips the *same* bit of the recovered plaintext, with no key needed:

$$(C_i \oplus \Delta) \oplus KS_i = P_i \oplus \Delta.$$

An attacker who knows the message format can thus make targeted edits the decryptor cannot notice. Confidentiality is not integrity. Real systems therefore use *authenticated encryption*, which bolts a message authentication code onto a mode (AES-GCM, ChaCha20-Poly1305) so any change is rejected — the subject of its own section.

2.6 Symbols

No new persistent symbols. Local to this section:

Local symbol	Meaning (this section)
E_k, D_k	block-cipher permutation and its inverse, on b-bit blocks
P_i, C_i	i-th b-bit plaintext / ciphertext block
n	number of blocks after padding
IV	initialization vector: random, non-secret, fresh per message
nonce	"number used once"; with i forms CTR's per-block input
KS_i	i-th keystream block, $E_k(\text{nonce} \,\|\| \, i)$ (CTR)
$\oplus$	bitwise XOR
Δ	an attacker's chosen bit-flip mask
i	block index, $1, \ldots, n$

3 Authenticated encryption

The modes of §2 hide a message but do nothing to stop *tampering* — flip a ciphertext bit and the plaintext changes, undetected (§7). *Authenticated encryption with associated data* (AEAD) adds integrity.

3.1 Two goals

- *confidentiality* — an eavesdropper learns nothing (what §2 gave);
- *integrity / authenticity* — tampering is detected, and the message came from a key-holder.

Encryption alone gives only the first.

3.2 The tool: a MAC

A *MAC* (message authentication code) stamps a message with a short *tag*, checked by recompute-and-compare (in *constant time*, so timing leaks nothing):

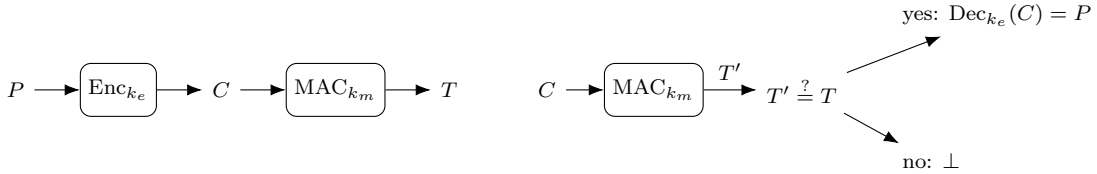
$$T = \mathrm{MAC}_{k_m}(M) \in \{0,1\}^t \ (t = 128), \qquad \text{accept} \iff \mathrm{MAC}_{k_m}(M) = T.$$

Only a holder of k_m can forge a valid tag, and one flipped bit breaks it. (HMAC, GHASH, and Poly1305 are MACs.)

3.3 Order matters

Three ways to combine a cipher and a MAC, with *independent* keys $k_e \neq k_m$:

- **Encrypt-and-MAC** (SSH): $C = \mathrm{Enc}_{k_e}(P)$, $T = \mathrm{MAC}_{k_m}(P)$ — tag is over the plaintext (leaks equality), and you must decrypt to verify.
- **MAC-then-Encrypt** (old TLS): $T = \mathrm{MAC}_{k_m}(P)$, $C = \mathrm{Enc}_{k_e}(P \parallel T)$ — must decrypt attacker data *before* checking T, the opening for padding oracles (§17).
- **Encrypt-then-MAC** (IPsec, recommended): $C = \mathrm{Enc}_{k_e}(P)$, $T = \mathrm{MAC}_{k_m}(C)$ — verify T on C first; a bad tag is $\perp$, no decryption.

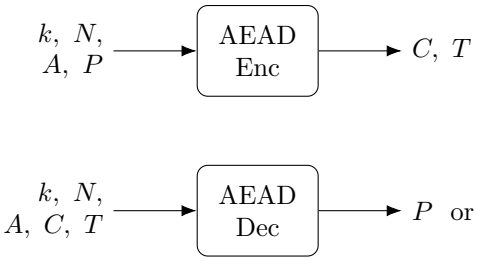


Encrypt-then-MAC (left) and its verify (right): the tag is checked on C before any decryption.

3.4 The AEAD interface

One primitive bakes in a correct composition, so application code cannot pick the wrong order or reuse a key:

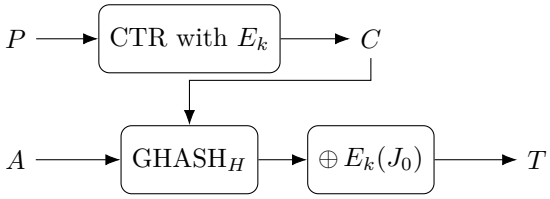
$$(C, T) = \mathrm{Enc}_k(N, A, P), \qquad \mathrm{Dec}_k(N, A, C, T) \in \{\, P,\ \perp \,\}.$$



- N — nonce, unique per key (§1); A — associated data: authenticated but *not* encrypted (e.g. a header);
- T — tag, covering *both* C and A; $\perp$ — "reject." Dec verifies first, so a forgery never gets decrypted.

3.5 AES-GCM

CTR (confidentiality, §2) + GHASH (a universal hash over GF(2^{128}), keyed by $H = E_k(0^{128})$; detailed in §4).



GHASH folds the blocks X_i of $A \parallel C \parallel \text{len}$ Horner-style, then is masked into the tag (J_0 = initial counter from N):

$$Y_0 = 0, \qquad Y_i = (Y_{i-1} \oplus X_i) \cdot H, \qquad T = Y_{\text{last}} \oplus E_k(J_0).$$

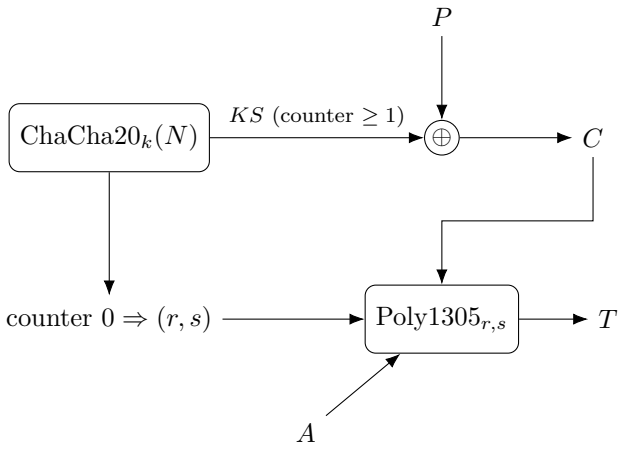
Send $N \parallel A \parallel C \parallel T$. The receiver recomputes T' from A, C only (never P); if $T' \neq T \to \perp$, else CTR-decrypt — verify-before-decrypt.

- The mask $\oplus E_k(J_0)$ is essential: GHASH alone is forgeable; a fresh-per-nonce mask makes the tag a MAC.
- **Nonce reuse is doubly fatal** — it breaks CTR confidentiality *and* leaks H, enabling forgery.

Fast with AES-NI / PCLMULQDQ hardware; NIST SP 800-38D.

3.6 ChaCha20-Poly1305

ChaCha20 (§1) + Poly1305. ChaCha20's keystream comes in counted 64-byte blocks: block 0 gives the *one-time* key (r, s) (its first 32 bytes — r = bytes 0–15, s = 16–31), blocks ≥ 1 give the keystream.



Poly1305 folds the 16-byte blocks c_i of $A \parallel C \parallel \text{len}$ Horner-style, mod $p = 2^{130} - 5$:

$$a_0 = 0, \qquad a_i = (a_{i-1} + c_i) \cdot r \bmod p, \qquad T = (a_{\text{last}} + s) \bmod 2^{128}.$$

The key (r, s) is used for *one* message (it is nonce-derived). Transmission and verify-before-decrypt are as in GCM: send $N \parallel A \parallel C \parallel T$, recompute the tag over A, C, then decrypt only on a match. No special hardware, constant-time — the pick without AES acceleration; RFC 8439.

	AES-GCM	ChaCha20-Poly1305
Cipher	AES (block, CTR)	ChaCha20 (stream)
MAC	GHASH, GF(2^{128})	Poly1305, mod $2^{130}-5$
Fastest with	AES-NI hardware	plain software
Nonce reuse	catastrophic	catastrophic
Standard	NIST SP 800-38D	RFC 8439

3.7 Symbols

No new persistent symbols. Local to this section:

Local symbol	Meaning (this section)
$\text{Enc}_{k_e}, \text{Dec}_{k_e}$	confidentiality-only encrypt / decrypt
MAC_{k_m}	message authentication code; tag $T = \text{MAC}_{k_m}(M)$
$k_e,\ k_m$	independent encryption and MAC keys ($k_e \neq k_m$)
N	nonce: unique per key
A	associated data: authenticated, not encrypted
$T,\ T'$	tag; and the tag recomputed at verification
$\perp$	"reject": returned when the tag fails
H	GCM's GHASH key, $H = E_k(0^{128})$
J_0	GCM's initial counter block, derived from N
$X_i,\ Y_i$	GHASH input blocks and running value
(r, s)	Poly1305 one-time key, from ChaCha20 block 0
$c_i,\ a_i$	Poly1305 input blocks and running value

4 Inside GHASH

AES-GCM (§3) authenticates with GHASH. HMAC got its own treatment (in the hashing collection); GHASH deserves one too. It is a *universal hash* turned into a MAC by a one-time mask — the *Wegman–Carter* recipe, which Poly1305 also follows.

4.1 A finite field: $\mathrm{GF}(2^{128})$

A *field* is a set you can add, subtract, multiply, and divide in (by anything nonzero) — like the rationals or reals, but here *finite*. We have used one already: the integers mod a prime p, written $\mathrm{GF}(p)$, where every nonzero value has an inverse (§8, §18).

$\mathrm{GF}(2^n)$ is a field with 2^n elements — but *not* the integers mod 2^n (that is not a field: 2 has no inverse mod 4). Instead each element is an n-bit string, read as a polynomial with $\{0, 1\}$ coefficients:

- **add** = XOR (add coefficients mod 2, no carries);
- **multiply** = multiply the polynomials, then reduce modulo a fixed *irreducible* degree-n polynomial — the field's "modulus," playing the role the prime p does in $\mathrm{GF}(p)$.

Because that modulus is irreducible, every nonzero element has an inverse, so division works.

	$\mathrm{GF}(p)$	$\mathrm{GF}(2^n)$
Elements	$0, \ldots, p-1$	n-bit strings (polynomials)
Add	mod-p addition	XOR
Multiply	mod-p multiply	carry-less multiply, then reduce
Used in	DH, ECDSA (mod p/n)	AES (2^8), GHASH (2^{128})

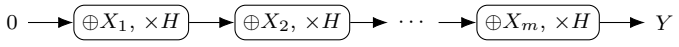
GHASH works in $\mathrm{GF}(2^{128})$ (16-byte blocks); AES (§6) uses $\mathrm{GF}(2^8)$ (bytes). The 2 means bits; the exponent, how many. CPUs have a carry-less multiply instruction (PCLMULQDQ) for it.

4.2 GHASH = a polynomial at a secret point

The key is $H = E_k(0^{128})$ (the block cipher on the all-zero block). The data blocks $X_1, \ldots, X_m$ (of $A \parallel C \parallel \text{len}$) are the coefficients of a polynomial, evaluated at H:

$$\mathrm{GHASH}_H(X_1, \ldots, X_m) = X_1 H^m \oplus X_2 H^{m-1} \oplus \cdots \oplus X_m H \qquad (\text{in } \mathrm{GF}(2^{128})).$$

In practice it is computed Horner-style, one block at a time:



$$Y_0 = 0, \qquad Y_i = (Y_{i-1} \oplus X_i) \cdot H, \qquad Y = Y_m = \mathrm{GHASH}_H(X).$$

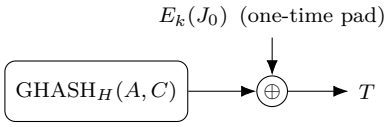
4.3 A universal hash is not yet a MAC

GHASH guarantees only that two *different* messages rarely collide — it is not, by itself, unforgeable. Output $\mathrm{GHASH}_H(M)$ as the tag directly and an attacker can forge: the output is a fixed polynomial in H, so they can shift the message and predict the change. So a universal hash needs a *secret, one-time pad* to become a MAC.

4.4 Wegman–Carter: mask with a one-time pad

GCM masks the GHASH output with a fresh per-nonce pad $E_k(J_0)$ (J_0 = the nonce-derived counter):

$$T = \mathrm{GHASH}_H(A, C) \oplus E_k(J_0).$$



The pad hides the universal-hash output, so forgery would require guessing it — infeasible. Universal hash $\oplus$ one-time pad is the *Wegman–Carter* MAC.

4.5 Why nonce reuse recovers H

The pad must be fresh. Reuse a nonce and it repeats, so two tags over messages with blocks X, X' cancel it:

$$T \oplus T' = \mathrm{GHASH}_H(X) \oplus \mathrm{GHASH}_H(X'),$$

a polynomial equation of degree $\sim m$ in the single unknown H. Solve it (factor over $\mathrm{GF}(2^{128})$) to recover H — and with H, forge any message. This is the "leaks H" of §3: the one-time pad must never repeat, which is exactly why nonce uniqueness is load-bearing.

4.6 Poly1305 is the same recipe

Poly1305 (§3) is the same construction in a prime field: a polynomial universal hash keyed by r, masked by a one-time pad s, mod $2^{130} - 5$. GHASH and Poly1305 are siblings — Wegman–Carter polynomial MACs; GHASH suits hardware (carry-less multiply), Poly1305 suits plain software.

4.7 Symbols

No new persistent symbols. Local to this section:

Local symbol	Meaning (this section)
$\mathrm{GF}(2^{128})$	field of 128-bit blocks: $+$ is XOR, $\times$ is carry-less multiply
H	the GHASH key, $H = E_k(0^{128})$
$X_i,\ Y_i$	input blocks (of $A \parallel C \parallel \mathrm{len}$) and the running value
$E_k(J_0)$	the one-time pad (J_0 = nonce-derived counter)
Wegman–Carter	universal hash $\oplus$ one-time pad = a MAC

5 Inside ChaCha20

§1 treated the stream cipher's keystream generator G as a black box, and §3 leaned on ChaCha20's *block counter* without saying what it is. Now we open the box. ChaCha20 (Bernstein, a refinement of Salsa20) is an *ARX* cipher: it builds its keystream with nothing but **A**dd, **R**otate, and **X**OR.

5.1 The state: a 4×4 grid of words

At heart is a *block function*: given the key, the nonce, and a block counter, it emits one 64-byte keystream block. It works on a state of 16 *words* (1 word = 32 bits), laid out as a 4×4 matrix — $16 \times 32 = 512$ bits = 64 bytes, exactly one block.

The four rows hold four different things:

c_0	c_1	c_2	c_3	4 constants ("`expand 32-byte k`")
k_0	k_1	k_2	k_3	256-bit key k (8 words)
k_4	k_5	k_6	k_7	
t	n_0	n_1	n_2	counter t | 96-bit nonce (3 words)

The top row is four fixed constants, the same in every ChaCha20 run — and not random-looking magic numbers. They are literally the 16 ASCII bytes of the text `expand 32-byte k` (a 16-character string = 16 bytes = 4 words of 32 bits). Cut the string into four 4-character pieces; each piece, read as a word, is one constant (its bytes packed little-endian, so `expa` becomes `0x61707865`):

	c_0	c_1	c_2	c_3
bytes	`expa`	`nd 3`	`2-by`	`te k`
word	`0x61707865`	`0x3320646e`	`0x79622d32`	`0x6b206574`

Why text? The constant is there as a fixed value that breaks the state's symmetry. And choosing it as plain readable text is a *nothing-up-my-sleeve* move: a carefully picked "magic" number could conceal a backdoor, but obvious ASCII — which even spells out that this is the 32-byte-key variant — leaves nowhere to hide one.

The counter t sits in word 12 — this is exactly the block counter of §3. (This is the RFC 8439 layout: a 32-bit counter and a 96-bit nonce.)

Three different time-scales share this one state. The key k is long-lived — reused across many messages. The nonce $n_0 n_1 n_2$ is drawn *fresh for each message* and held fixed while that message is encrypted. The counter t then runs $0, 1, 2, \ldots$ over the 64-byte blocks *within* that single message.

So within one message only t moves: bump it and the whole block changes, giving the next 64 bytes of keystream, while the nonce and key stay put. For the *next* message you draw a new nonce and reset t — the key usually does *not* change, yet the nonce changes anyway. Its freshness is not tied to the key; it must differ every message. That is the (key, nonce)-uniqueness rule of §1 and §3: never let a (key, nonce) pair repeat.

5.2 The quarter-round

All the mixing is done by one primitive, the *quarter-round* QR — a small function of *four* words. A call $\mathrm{QR}(a, b, c, d)$ takes four of the state's words and updates them in place. The names a, b, c, d are placeholders for "the four words this call happens to work on"; which of the state words $x_0, \ldots, x_{15}$ they stand for depends on where QR is applied — fixed by the column/diagonal pattern below. (They are generic argument names, *not* the constants $c_0 \ldots c_3$ from above.)

Its four words are updated by these eight steps, run *strictly in sequence* from top to bottom — each step reads the words the steps above it just wrote, so nothing here is parallel:

1. $a \leftarrow a \boxplus b$
2. $d \leftarrow (d \oplus a) \lll 16$
3. $c \leftarrow c \boxplus d$
4. $b \leftarrow (b \oplus c) \lll 12$
5. $a \leftarrow a \boxplus b$
6. $d \leftarrow (d \oplus a) \lll 8$
7. $c \leftarrow c \boxplus d$
8. $b \leftarrow (b \oplus c) \lll 7$

For example, step 2 reads the *new* a that step 1 just wrote, and step 3 the new d from step 2 — the updates chain.

Here $\boxplus$ is addition mod 2^{32}, $\oplus$ is XOR, and $w \lll k$ is a left-rotation of the 32-bit word w by k bits. That is the whole trio: add (mixes across bit positions via carries), rotate (moves bits between positions), XOR (the cheap, reversible combiner).

As a concrete binding: the first column quarter-round (below) sets $a = x_0,\ b = x_4,\ c = x_8,\ d = x_{12}$, while a diagonal one sets $a = x_0,\ b = x_5,\ c = x_{10},\ d = x_{15}$ — the same recipe, applied to different words.

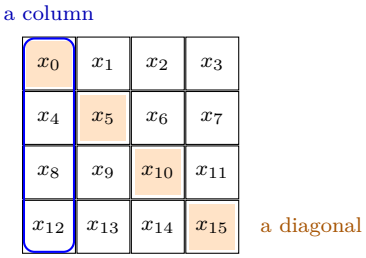
There are no S-boxes and no lookup tables — a deliberate choice we return to in §5 below.

The same quarter-round powers the BLAKE3 hash (in the hashing collection); BLAKE3's compression function is this mix run with right-rotations and two message words folded in.

5.3 Rounds: columns, then diagonals

One quarter-round touches only four words. To mix all 16, ChaCha alternates between the *columns* and the *diagonals* of the grid. A *double round* is a column round *followed by* a diagonal round (both, in that order — not a choice between them):

column round:	$\mathrm{QR}(x_0, x_4, x_8, x_{12}),\ \mathrm{QR}(x_1, x_5, x_9, x_{13}),\ \mathrm{QR}(x_2, x_6, x_{10}, x_{14}),\ \mathrm{QR}(x_3, x_7, x_{11}, x_{15});$
diagonal round:	$\mathrm{QR}(x_0, x_5, x_{10}, x_{15}),\ \mathrm{QR}(x_1, x_6, x_{11}, x_{12}),\ \mathrm{QR}(x_2, x_7, x_8, x_{13}),\ \mathrm{QR}(x_3, x_4, x_9, x_{14}).$



The name says the count: **ChaCha20** repeats this fixed pair 10 times — column, diagonal, column, diagonal, … — for 20 rounds in all. The order never varies, and nothing here is random.

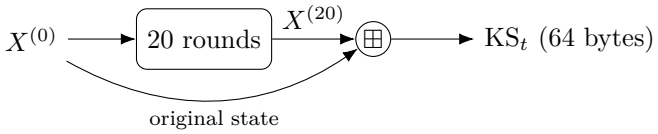
It *must* be deterministic: decryption regenerates the identical keystream from the same key, nonce, and counter (§1), so both sides have to walk exactly the same rounds. The only per-message freshness is the nonce, which is sent in the clear — not a secret random choice.

Diffusion compounds fast — after a couple of double rounds every output word depends on every input word.

5.4 Feed-forward, then the keystream

Call the starting matrix $X^{(0)}$ and the matrix after 20 rounds $X^{(20)}$. The block function does not output $X^{(20)}$ directly; it first adds the *original* state back in, word by word (mod 2^{32}):

$$\mathrm{KS}_t = X^{(20)} \boxplus X^{(0)}.$$



That feed-forward is what makes the block function one-way: the 20 rounds alone are a reversible permutation, but adding $X^{(0)}$ back means you cannot run it backwards to recover the key or counter from the keystream — the same trick as SHA-2's Davies–Meyer (in the hashing collection).

The result KS_t is the 64-byte keystream block for counter t. To encrypt, do what §1 and §2 described: generate $\mathrm{KS}_0, \mathrm{KS}_1, \ldots$ by incrementing t, and XOR them onto the plaintext.

This is the missing piece from §3: in ChaCha20-Poly1305 the counter-0 block supplies the Poly1305 key (its first 32 bytes) and the message rides on counters $1, 2, \ldots$.

5.5 Why ARX

The all-ARX design buys two things. Because every step is arithmetic on whole words — never a table indexed by secret data — ChaCha20 runs in *constant time*, immune to the cache-timing leaks that dog naive S-box ciphers.

And ARX is cheap and simple on any CPU, with no special hardware needed — which is precisely why ChaCha20-Poly1305 (§3) is the pick on phones and other machines without AES acceleration.

The 20 rounds are a generous margin: the best known attacks reach only about 7.

5.6 Symbols

No new persistent symbols. Local to this section:

Local symbol	Meaning (this section)
word	32 bits
$x_0, \ldots, x_{15}$	the 16-word (4×4) ChaCha20 state
$c_0, \ldots, c_3$	the four fixed constants ("`expand 32-byte k`")
$k_0, \ldots, k_7$	the eight words of the 256-bit key k
t	block counter (state word 12)
n_0, n_1, n_2	the three words of the 96-bit nonce
QR	the quarter-round (mixes four state words)
$\boxplus$	addition mod 2^{32} (word-wise)
$\lll k$	left-rotation of a 32-bit word by k bits
$X^{(0)}, X^{(20)}$	the state before / after the 20 rounds
KS_t	the 64-byte keystream block for counter t

6 Inside AES

§5 built a cipher from pure ARX. AES (Rijndael, the NIST standard since 2001) takes the other classic route: a *substitution–permutation network* (SP-network) — alternating a byte-substitution layer with a byte-shuffling layer, interleaved with key XORs.

Unlike ChaCha20, AES is a *block cipher* (§1): it maps one 128-bit block to one 128-bit block under the key, and to encrypt a real message you wrap it in a mode (§2).

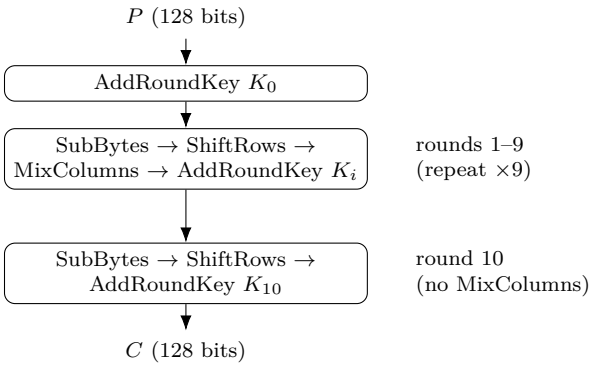
6.1 The state and the rounds

AES works on a *state*: the 128-bit block written as a 4×4 grid of *bytes* (16 bytes = 128 bits). Write $s_{r,c}$ for the byte in row r, column c ($r, c \in \{0, 1, 2, 3\}$). The input bytes fill the grid down columns — bytes 0–3 are column 0, bytes 4–7 column 1, and so on.

$s_{0,0}$	$s_{0,1}$	$s_{0,2}$	$s_{0,3}$
$s_{1,0}$	$s_{1,1}$	$s_{1,2}$	$s_{1,3}$
$s_{2,0}$	$s_{2,1}$	$s_{2,2}$	$s_{2,3}$
$s_{3,0}$	$s_{3,1}$	$s_{3,2}$	$s_{3,3}$

16 bytes, 4×4 (= 128 bits)

A 128-bit key gives $N_r = 10$ rounds (a 192-bit key gives 12, a 256-bit key 14). Each full round is four steps in fixed order — **SubBytes**, **ShiftRows**, **MixColumns**, **AddRoundKey** — with one twist at the ends:



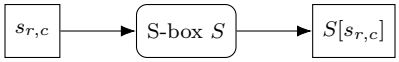
An extra AddRoundKey runs *before* round 1 (so no plaintext block is ever sent through the rounds unkeyed), and the *final* round drops MixColumns.

Three of the four steps — SubBytes, ShiftRows, MixColumns — are fixed, public, key-free shuffles. Only AddRoundKey injects the secret. The security comes from repeating the substitute-shuffle-mix-key alternation enough times that the two are hopelessly entangled.

6.2 SubBytes — the substitution (confusion)

Each byte is replaced by another via a fixed 256-entry lookup table, the *S-box* S:

$$s_{r,c} \leftarrow S[\, s_{r,c}\,] \qquad \text{for all } r, c.$$



This is the cipher's *only* nonlinear step, and it supplies *confusion* (Shannon's term): it makes the relationship between key and ciphertext intricate, so the output cannot be unpicked with linear algebra. The S-box itself is built from the multiplicative inverse in $\mathrm{GF}(2^8)$ followed by a fixed affine map — chosen for strong nonlinearity — where $\mathrm{GF}(2^8)$ is byte arithmetic (each byte a polynomial; add = XOR, multiply = carry-less multiply reduced by a fixed degree-8 polynomial).

6.3 ShiftRows — shuffle across columns (diffusion)

Row r is cyclically shifted left by r bytes: row 0 stays put, row 1 moves one place, row 2 two, row 3 three. (Cells show the byte index, numbered down columns.)

0	4	8	12
1	5	9	13
2	6	10	14
3	7	11	15

before

→

0	4	8	12	
5	9	13	1	shift 1
10	14	2	6	shift 2
15	3	7	11	shift 3

after

On its own this just moves bytes around. Its job is to feed MixColumns: after the shift, each column holds bytes that started in four *different* columns, so the next step mixes bytes from all over the block — the start of *diffusion* (spreading each input's influence widely).

6.4 MixColumns — mix within a column (diffusion)

Each column is treated as a 4-byte vector and multiplied by a fixed matrix over $\mathrm{GF}(2^8)$:

$$\begin{pmatrix} s'_{0,c} \\ s'_{1,c} \\ s'_{2,c} \\ s'_{3,c} \end{pmatrix} = \begin{pmatrix} 2 & 3 & 1 & 1 \\ 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 3 \\ 3 & 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} s_{0,c} \\ s_{1,c} \\ s_{2,c} \\ s_{3,c} \end{pmatrix} \qquad (c = 0, \ldots, 3).$$

Spelling out the first row, with $\cdot$ a $\mathrm{GF}(2^8)$ multiply and $\oplus$ XOR:

$$s'_{0,c} = (2 \cdot s_{0,c}) \oplus (3 \cdot s_{1,c}) \oplus s_{2,c} \oplus s_{3,c}.$$

The constants $1, 2, 3$ are just small field elements. ShiftRows and MixColumns together give full diffusion: after about two rounds every output byte depends on every input byte. (The final round omits MixColumns — at the very end it adds no security, and leaving it out keeps encryption and decryption structurally aligned.)

6.5 AddRoundKey and the key schedule

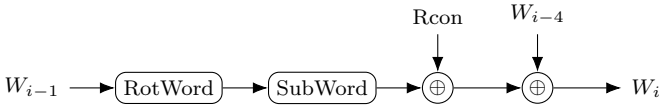
AddRoundKey XORs the round key K_i (also 128 bits, a 4×4 byte grid) into the state, byte for byte:

$$s_{r,c} \leftarrow s_{r,c} \oplus (K_i)_{r,c}.$$

This is the only step that touches the secret. XOR is its own inverse, so decryption removes the same key with another XOR.

The $N_r + 1 = 11$ round keys are not 11 independent secrets — they are stretched from the one key by the *key schedule*. Take the key as four 4-byte words $W_0, \ldots, W_3$; the rest follow a recurrence. For

most words $W_i = W_{i-4} \oplus W_{i-1}$; but every fourth word first passes W_{i-1} through a small function g (rotate its bytes, push them through the S-box, then XOR a round constant Rcon):



Round key K_i is the four words $W_{4i}, \ldots, W_{4i+3}$. Feeding the words through the S-box and the changing Rcon makes the round keys differ unpredictably, so learning one tells an attacker little about the others.

6.6 SP-network vs. ARX

AES and ChaCha20 (§5) reach the same goal by opposite means: AES leans on a lookup-table S-box plus linear mixing, ChaCha20 on add/rotate/XOR with no tables at all.

That S-box is the catch. Implemented naively as a table indexed by secret bytes, it can leak through cache-timing; this is exactly the weakness ChaCha20's table-free design avoids. AES's answer is hardware: the **AES-NI** instructions do a whole round at once, making AES both very fast *and* constant-time on modern CPUs.

That split is why §3's two AEADs coexist: AES-GCM wins where AES-NI is present, ChaCha20-Poly1305 where it is not.

	AES	ChaCha20
Type	block cipher	stream cipher
Design	SP-network (S-box)	ARX (no tables)
Unit	byte (GF(2^8))	32-bit word
Nonlinearity	the S-box	carry in $\boxplus$
Fastest with	AES-NI hardware	plain software

6.7 Symbols

No new persistent symbols. Local to this section:

Local symbol	Meaning (this section)
$s_{r,c}$	state byte at row r, column c; the state is 4×4 bytes = 128 bits
N_r	number of rounds (10 for AES-128)
$S[\cdot]$	the S-box: a fixed 256-entry byte substitution
K_i	the i-th round key (128 bits), $i = 0, \ldots, N_r$
GF(2^8)	byte field: add = XOR, multiply = carry-less mod a fixed polynomial
W_i	key-schedule words (4 bytes each); $K_i = (W_{4i}, \ldots, W_{4i+3})$
RotWord, SubWord, Rcon	key-schedule helpers (rotate, S-box, round constant)
$\oplus$	bitwise XOR

7 Nonce management

§1, §2, and §3 all ended with the same warning: never reuse a (key, nonce) pair. This section is about how to actually *guarantee* that — and what to do when you cannot.

7.1 Why a repeated nonce is a disaster

Reuse breaks things in two separate ways.

Confidentiality. A repeated (key, nonce) gives the *same* keystream KS. Two messages masked with it satisfy

$$C \oplus C' = (P \oplus KS) \oplus (P' \oplus KS) = P \oplus P',$$

so the keystream cancels and the attacker learns $P \oplus P'$ — the two-time-pad break of §1.

Authenticity. In GCM (§3) a repeat is worse still: it lets the attacker solve for the GHASH key H and then *forge* tags for messages of their choosing. So a single slip turns into a total break, not just a leak.

That is why nonce uniqueness is load-bearing for the whole construction — and why it deserves real care.

7.2 Why $P \oplus P'$ is already a break

It is fair to object that $P \oplus P'$ looks useless: to recover P you would XOR by P', which you do not have. And that is right — *if the plaintexts were random*, $P \oplus P'$ would leak nothing. The break works because real plaintexts are *not* random: they are full of structure — known headers, fixed fields, ordinary words, predictable formatting.

That redundancy is the lever. Suppose you can *guess* a stretch of one message; call the guess g. XOR it into $P \oplus P'$:

$$(P \oplus P') \oplus g \;=\; P' \qquad \text{wherever } g = P.$$

A correct guess of part of P immediately hands you the same part of P' (and vice versa). This is the $P \oplus P' \oplus P' = P$ you spotted — except you supply a *guess* of one plaintext, not the unknown one.

Guesses come from *cribs*: a likely word or a known field. Slide a candidate — a common word like `the`, or a protocol header — along $P \oplus P'$ at each offset; a wrong position yields garbage in the other message, a right one yields readable text. Natural-language redundancy makes the right alignment stand out, and each recovered fragment becomes a crib for more. This is *crib dragging*, and it peels apart both messages.

Even with no crib to start, footholds are everywhere: $P \oplus P'$ is *zero* at every position where the two messages agree (a shared header or greeting appears as a run of zero bytes), and a space (`0x20`) XORed against a letter merely flips its case bit — a statistical tell that exposes where the spaces sit in each message.

So one keystream reuse downgrades "perfectly secret" to "two ciphertexts whose XOR is the XOR of two *predictable* messages" — routinely solvable. The one-time pad earns its name literally: it is secure only when the pad is used *once*.

7.3 Two ways to keep nonces unique

There are two strategies, with opposite failure modes.

Counter nonces (stateful): hold a counter and increment it for every message, never repeating a value. This gives the largest message budget — a 96-bit counter never wraps in practice. The danger is the *state*: a reboot, a restored backup, or a cloned VM can rewind the counter and silently reissue a value.

Random nonces (stateless): draw a fresh random nonce each time. There is no state to lose. The danger is *chance*: two messages may happen to draw the same nonce — the birthday problem, next.

7.4 The birthday bound on random nonces

With n-bit random nonces, after q messages under one key the chance that two coincide is about

$$\Pr[\text{collision}] \;\approx\; \frac{q^2}{2^{n+1}}.$$

To keep that negligible (say below 2^{-32}), you need $q \lesssim 2^{(n-32)/2}$. That ceiling is lower than people expect:

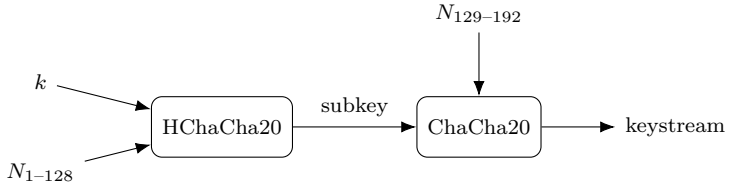
Nonce size	Used by	Random nonces safe up to
96 bits	GCM, ChaCha20 (§3)	$\approx 2^{32}$ messages
192 bits	XChaCha20 (below)	$\approx 2^{80}$ messages

So a 96-bit random nonce is fine for billions of messages but not for an unbounded stream; past the limit you must rekey — or use a bigger nonce.

7.5 Bigger nonces: XChaCha20

The clean fix for random nonces is to make the nonce so large that collisions never happen. XChaCha20 extends ChaCha20's 96-bit nonce to 192 bits.

It does so in two stages. First it runs *HChaCha20* — the ChaCha core (§5) used as a key-derivation step — on the key and the *first* 128 bits of the nonce, giving a fresh 256-bit subkey. Then it runs ordinary ChaCha20 under that subkey with the remaining 64 nonce bits.

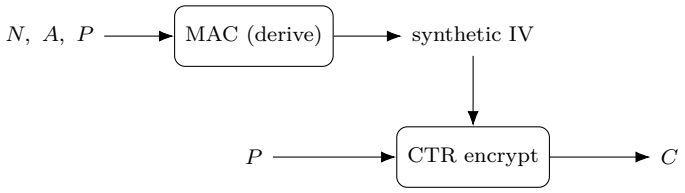


The upshot is a 192-bit nonce, whose birthday bound ($\approx 2^{80}$ messages) is so far off that you can pick nonces at random and never think about collisions again. (XSalsa20 does the same for Salsa20.)

7.6 Surviving reuse: misuse-resistant AEAD

A different philosophy: rather than *prevent* reuse, limit the damage when it happens. That is what AES-GCM-SIV does (SIV = *synthetic IV*).

Its trick is to derive the encryption IV from a MAC of the nonce, the associated data, *and the plaintext*, then CTR-encrypt with that IV (which also serves as the tag).



Because the IV depends on the message content, two *different* messages under the same (key, nonce) get different IVs — hence different keystreams, and no two-time pad. The only thing a repeat leaks is whether the *exact same* message was sent twice (identical ciphertext). That is a mild, bounded leak next to GCM's catastrophe.

The cost: the IV must be computed from the whole plaintext before encryption can start, so AES-GCM-SIV needs two passes and cannot encrypt a stream on the fly. You trade a little speed for robustness against a footgun.

7.7 Which to use

Situation	Use
Reliable counter state	a counter nonce (largest budget)
Stateless or many senders	random nonces — prefer 192-bit (XChaCha20)
Uniqueness can't be guaranteed	a misuse-resistant AEAD (AES-GCM-SIV)

Underneath all three, the rule never changes: a (key, nonce) pair must not repeat. When you are unsure whether it might, rekey.

7.8 Symbols

No new persistent symbols. Local to this section:

Local symbol	Meaning (this section)
N	the nonce
n	nonce length in bits
q	number of messages encrypted under one key
KS	keystream (§1)
P'	a second plaintext sent under the same (key, nonce)
g	a guessed plaintext fragment (a "crib")
subkey	the 256-bit ChaCha20 key HChaCha20 derives (XChaCha20)
synthetic IV	an IV derived from the message itself (AES-GCM-SIV)

8 Diffie–Hellman key exchange

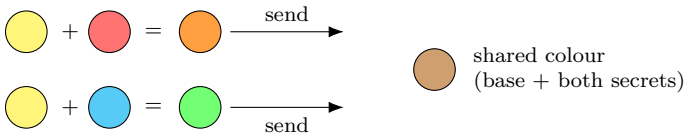
Every section so far opened with "a shared secret key k" and simply assumed both sides had it. But how do two parties who have never met agree on k over a channel an eavesdropper can read? You cannot just send k — the eavesdropper would see it.

Diffie–Hellman (1976) solves exactly this. It is the first piece of *public-key* crypto in this document: a way for two parties to agree on a shared secret in the open, where the eavesdropper sees *every* message exchanged and still cannot compute the secret.

8.1 The idea: mixing that will not un-mix

The trick is an operation that is easy to do but hard to undo. The classic intuition is *paint*: mixing two colours is easy, but separating a mixture back into its ingredients is practically impossible.

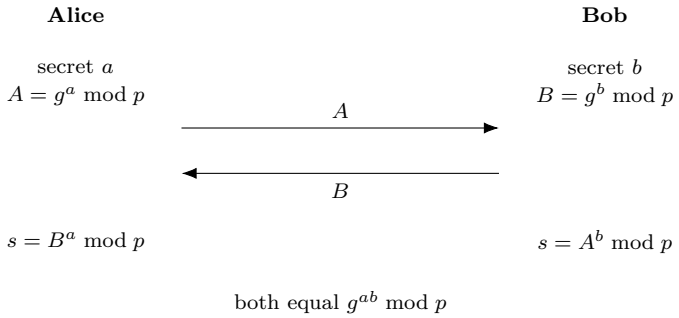
Both parties start from a public base colour. Each keeps a secret colour and mixes it into the base; they swap the resulting mixtures in the open. Each then stirs in their own secret again — and both arrive at the *same* final colour (base + both secrets), while an eavesdropper, holding only the two public mixtures, cannot un-mix them to reach it.



The mathematical "mixing" is *modular exponentiation*. Fix a large public prime p and a public base g. Raising g to a power mod p is easy; but recovering the exponent from the result — the *discrete logarithm* — is believed infeasible for large p. That one-wayness is what the eavesdropper runs into.

8.2 The exchange

Public, known to everyone: the prime p and the base g. Each party draws one private exponent.



Both land on the same value because the exponents just multiply:

$$B^a = (g^b)^a = g^{ab} = (g^a)^b = A^b \pmod{p}.$$

The eavesdropper sees g, p, $A = g^a$, and $B = g^b$, but to get g^{ab} they would need a or b — i.e. solve a discrete logarithm. That is the *Diffie–Hellman problem*, believed hard for well-chosen large p (today, 2048 bits or more).

Notation flag. The modulus here is the lower-case prime p, *not* the plaintext P used elsewhere in this document.

8.3 A worked example

Small numbers make it concrete. Take public $p = 23$, $g = 5$.

- Alice picks $a = 6$ and sends $A = 5^6 \bmod 23 = 8$.
- Bob picks $b = 15$ and sends $B = 5^{15} \bmod 23 = 19$.
- Alice computes $s = B^a = 19^6 \bmod 23 = 2$.
- Bob computes $s = A^b = 8^{15} \bmod 23 = 2$.

Both get $s = 2$, which equals $g^{ab} = 5^{90} \bmod 23 = 2$. An eavesdropper holds $g = 5$, $p = 23$, $A = 8$, $B = 19$ — and to find a must solve $5^a \equiv 8 \pmod{23}$ by discrete log. At 23 that is a quick search; at 2048 bits it is hopeless.

8.4 From shared secret to key

The shared s is a big number, not yet a key. You do not use it directly: run it through a key-derivation function (a hash-based KDF, from the hashing collection) to get a uniform symmetric key

$$k = \mathrm{KDF}(s).$$

From there, everything in §1–§7 applies — k feeds AES-GCM or ChaCha20-Poly1305 as usual. Diffie–Hellman is the missing first step that produces the k the rest of the document assumed.

8.5 The catch: it does not authenticate

Plain Diffie–Hellman defeats a *passive* eavesdropper, but not an *active* attacker. Nothing in the exchange proves *who* is on the other end.

A *man-in-the-middle*, Mallory, can run a separate exchange with each side — agreeing on a secret s_1 with Alice and a different s_2 with Bob — then sit in the middle, decrypting with one key and re-encrypting with the other, reading everything. Neither side notices.

So the exchange must be *authenticated*: the public values are tied to an identity by a signature, a certificate, or a pre-shared key. TLS, for instance, has the server sign its Diffie–Hellman value. Diffie–Hellman gives secrecy of the agreement; proving who you agreed *with* is a separate, mandatory job.

8.6 In practice: ECDH and forward secrecy

Two refinements dominate real use.

Elliptic-curve Diffie–Hellman (ECDH) runs the same idea in a different group: points on an elliptic curve, where "raise to a power" becomes "multiply a point by a scalar." The hard problem is the same shape, but breaking it needs far bigger effort per bit — so a 256-bit curve key (e.g. X25519) matches a 3072-bit classical one. Smaller and faster, it is the modern default.

Forward secrecy comes from using *ephemeral* secrets — fresh a, b per session, discarded afterwards (the "E" in ECDHE). Then even if a party's long-term key is later stolen, past sessions stay secret: their one-time exponents are already gone.

8.7 Symbols

No new persistent symbols. Local to this section:

Local symbol	Meaning (this section)
p	large public prime, the modulus (*not* the plaintext P)
g	public base (generator)
$a,\ b$	Alice's and Bob's private exponents (secret)
$A,\ B$	public values $g^a \bmod p$ and $g^b \bmod p$
s	the shared secret $g^{ab} \bmod p$
KDF	key-derivation function turning s into the key k

9 Digital signatures

§8 left a hole: Diffie–Hellman agrees a key but cannot prove *who* is on the other end, so a man-in-the-middle slips in. Closing that hole is what digital signatures do.

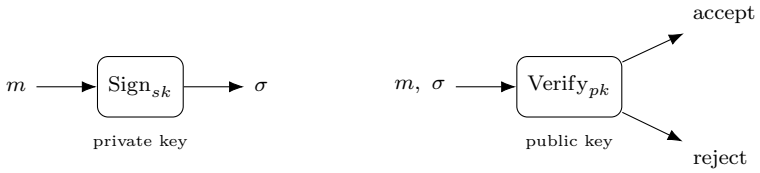
Could a MAC (§3) prove identity? No — a MAC needs a *shared* secret, and a shared secret is exactly what we have not got yet. We need to prove a message came from a specific party with *no* pre-shared secret, in a way *anyone* can check. That is a signature.

9.1 Key pairs: sign with private, verify with public

Each party holds a *key pair*: a *private key* sk kept secret, and a *public key* pk published to the world. They drive two operations:

$$\sigma = \mathrm{Sign}_{sk}(m), \qquad \mathrm{Verify}_{pk}(m, \sigma) \in \{\mathrm{accept}, \mathrm{reject}\}.$$

Here m is the message and σ its signature.



Only the holder of sk can produce a σ that verifies, and pk reveals nothing that helps forge one. So *anyone* can verify, but only *one* party can sign.

That asymmetry is the whole difference from a MAC:

	MAC (§3)	Signature
Keys	one shared secret	private (sign) + public (verify)
Who can sign	both parties	only the private-key holder
Who can verify	both parties	anyone with pk
Shared secret	required	none
Non-repudiation	no	yes

The last row matters: because only the signer holds sk, they cannot later deny a valid signature — *non-repudiation*, something a MAC (where both sides share the key) can never give.

9.2 Sign the hash, not the message

The signature math takes a fixed-size input, but messages are any length. So you hash first (*hash-then-sign*):

$$\sigma = \mathrm{Sign}_{sk}\big(H(m)\big),$$

where H is a collision-resistant hash (from the hashing collection). Collision resistance is essential: if $H(m) = H(m')$, one signature would be valid for both messages.

9.3 How it works: RSA and elliptic curves

Every scheme rests on a *trapdoor*: signing is easy with sk, forging is infeasible without it.

RSA (trapdoor: factoring is hard). The public key is a modulus $N = pq$ (product of two big primes) and an exponent e; the private key is d with $ed \equiv 1 \pmod{\varphi(N)}$. Signing applies the private exponent, verifying the public one, and they undo each other:

$$\sigma = H(m)^d \bmod N, \qquad \text{accept iff } \sigma^e \equiv H(m) \pmod{N}.$$

Forging needs d, which needs factoring N — infeasible for large N. (Real RSA pads the hash first, e.g. PSS; signing the bare hash is unsafe.)

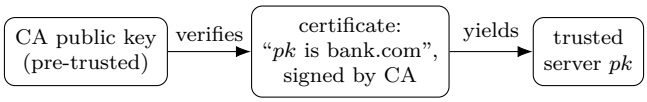
Elliptic curves (trapdoor: the curve discrete log of §8). The private key is a secret scalar, the public key a curve point. ECDSA and Ed25519 are the modern schemes — much smaller and faster than RSA for equal security.

A nonce footgun, again. ECDSA needs a fresh, unpredictable secret value for *each* signature. Reuse it — or let it be guessable — across two signatures and the private key falls straight out, the same nonce-reuse disaster as §7. It has broken real systems (the Sony PS3, early Bitcoin wallets). Ed25519 removes the footgun by deriving that value deterministically from the key and message.

9.4 Certificates: trusting a public key

A signature proves "whoever holds *sk* signed this" — but how do you know a given *pk* belongs to the *right* party (the real bank, not Mallory)?

A *certificate* answers that: a trusted *Certificate Authority* (CA) signs a statement binding a public key to an identity. You check that certificate with the CA's *own* public key, which you already trust because it ships with your OS or browser.



This is the *chain of trust*: trust in one pre-installed CA key extends, via signatures, to the public key of any site it vouches for.

9.5 Authenticating Diffie–Hellman

Now §8's man-in-the-middle is finished. Each party *signs* its Diffie–Hellman public value with its private key; the other verifies the signature with the partner's certificate-vouched public key.

Mallory cannot forge that signature, so she cannot substitute her own Diffie–Hellman value without being caught — the exchange is now *authenticated*. Key agreement (§8) plus signed identities is the combination every real secure channel is built from.

9.6 Symbols

No new persistent symbols. Local to this section:

Local symbol	Meaning (this section)
$sk,\ pk$	private (signing) key and public (verification) key
Sign, Verify	the signing and verification operations
σ	the signature
$m,\ H(m)$	the message and its hash (what actually gets signed)
$N,\ e,\ d$	RSA modulus $N = pq$, public exponent e, private exponent d
CA	Certificate Authority: vouches for a public key

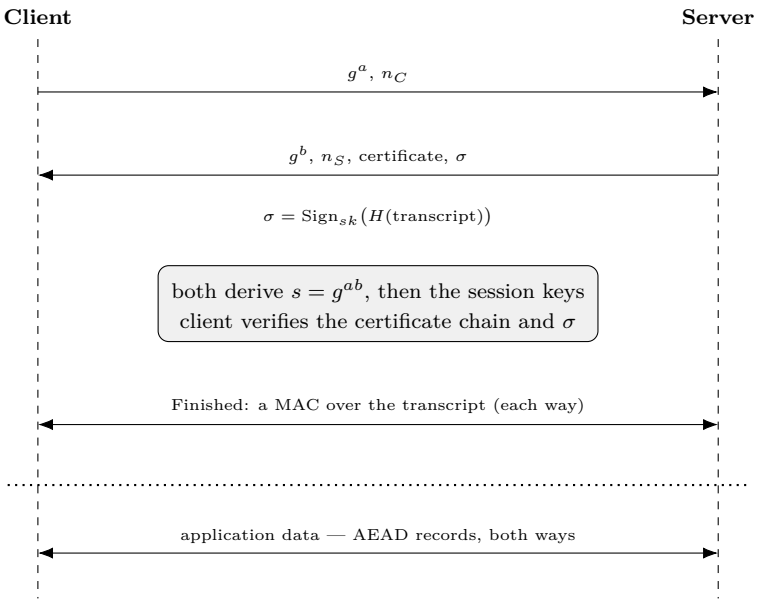
10 Putting it together: a secure channel

Every piece is now on the table:

- AEAD for confidentiality *and* integrity (§3), built on AES (§6) or ChaCha20 (§5);
- Diffie–Hellman to agree a key over a public channel (§8);
- signatures and certificates to prove identity (§9);
- nonce discipline to keep it all safe (§7).

A *secure channel* — TLS is the everyday example — bolts them into one protocol. Here is a simplified TLS-1.3-style flow.

10.1 The handshake



The client opens with an *ephemeral* Diffie–Hellman share g^a (§8) and a fresh random nonce n_C. "Ephemeral" means a is generated for this session and discarded after — that is what buys forward secrecy.

The server replies with its own share g^b, its *own* nonce n_S, its certificate (§9), and a signature σ. The two nonces are *different* — one chosen by each side — and both are folded into the key derivation, so the session keys draw on randomness from both parties (and a replayed handshake yields different keys).

What is that signature? Not encryption. It is

$$\sigma = \mathrm{Sign}_{sk}\big(H(\text{transcript})\big)\ :$$

the server signs a hash of the *entire handshake transcript so far* (both shares, both nonces, the certificate) with its *private* signing key (§9), and the client checks it with the public key in the certificate. Signing the whole transcript — not just g^b — binds the server's identity to *this* exact exchange, so a man-in-the-middle cannot lift the signature onto different values.

Both sides now compute the shared secret $s = g^{ab}$ (§8). Crucially, the client verifies the certificate chain (up to a trusted CA) and the signature *before* trusting s: Mallory cannot forge σ, so she cannot substitute her own g^b — the man-in-the-middle is shut out.

Finally each side sends a *Finished* message. It is not a signature but a MAC (§3): the KDF derives one more value from s — a separate *finished key* k_{fin} — and each side computes

$$\text{Finished} = \text{MAC}_{k_{\text{fin}}}\big(H(\text{transcript})\big),$$

an HMAC over the hash of all handshake messages so far. Each verifies the other's. A match proves two things at once: both sides derived the same secret (only a holder of k_{fin} could produce the tag) and both saw the identical, untampered handshake (any altered message changes $H(\text{transcript})$). Note the division of labour — the signature σ used the server's *private* key to prove *identity*; the Finished uses a *shared derived* key to prove *agreement*. Only then does application data flow.

10.2 Deriving the session keys

The raw s is never used directly. It is fed — with both nonces and the transcript — into a key-derivation function (HKDF, from the hashing collection), which emits *two different* keys, one for each direction:

$$\underbrace{k_{C\to S}}_{\text{client}\to\text{server}} \;,\; \underbrace{k_{S\to C}}_{\text{server}\to\text{client}} \;=\; \text{KDF}(s,\; n_C,\; n_S,\; \text{transcript}), \qquad k_{C\to S} \neq k_{S\to C}.$$

They are *not* the same key. Why two? If both directions shared one key and each began its counter nonce (§7) at zero, the client's first record and the server's first record would reuse the same (key, nonce) pair — the two-time-pad disaster of §7. A separate key per direction removes that risk entirely.

But *both* parties derive the same pair from s, so each holds *both* keys — a key names a *direction*, not an owner. The client encrypts its traffic under $k_{C\to S}$ and the server decrypts it under that *same* $k_{C\to S}$; the server's replies travel under $k_{S\to C}$, which the client uses to open them. AEAD is symmetric (§3) — the same key both seals and opens — so two shared keys are all it takes.

10.3 The data phase

From here every message — every *record* — is sealed with AEAD (§3) under the direction's key, using AES-GCM or ChaCha20-Poly1305.

Each record carries a counter nonce (§7) that increments per record: it never repeats, and it doubles as a sequence number, so a dropped, reordered, or replayed record is caught.

Record headers ride along as *associated data* (§3) — authenticated so they cannot be tampered with, but left readable because routing needs them.

10.4 Where each guarantee comes from

The point of the assembly is that each property of the channel traces to exactly one piece:

Guarantee	Comes from	Where
Confidentiality	AEAD encryption	§3, §6, §5
Integrity & authenticity (data)	the AEAD tag	§3
Shared key over a public channel	(ephemeral) Diffie–Hellman	§8
Identity / no man-in-the-middle	signatures + certificates	§9
Forward secrecy	ephemeral DH keys	§8
No nonce reuse	per-direction keys, counter nonces	§7

No single primitive is a secure channel on its own; the channel is what you get when they are composed in the right order, each covering the gap the previous one left.

10.5 Symbols

No new persistent symbols. Local to this section:

Local symbol	Meaning (this section)
g^a, g^b	the client's and server's ephemeral DH shares (§8)
s	the shared secret g^{ab} (§8)
$n_C,\ n_S$	the client's and server's (distinct) handshake nonces
σ	the server's signature, $\mathrm{Sign}_{sk}(H(\text{transcript}))$
transcript	all handshake messages exchanged so far
$k_{C\to S},\ k_{S\to C}$	the two *distinct* per-direction session keys
k_{fin}	a separate finished key (from s) that keys the Finished MAC
record	one AEAD-sealed message in the data phase

11 A post-quantum note

A large-scale quantum computer would not break cryptography evenly. The damage is lopsided — and which half falls is exactly the split this document has been built around.

11.1 What a quantum computer breaks

Shor's algorithm solves integer factoring and discrete logarithms efficiently on a quantum computer. Those are precisely the hard problems underneath the public-key half: factoring for RSA (§9), discrete log for Diffie–Hellman and ECDH (§8), and the curve discrete log for ECDSA/Ed25519 (§9). All of it — key agreement *and* signatures, hence the whole handshake of §10 — would fall completely.

Grover's algorithm only speeds up brute-force search quadratically (square-root). Against the symmetric world — AES (§6), ChaCha20 (§5), AEAD, hashes — it merely halves the effective strength, which you fix by doubling the key or output size. The symmetric core survives essentially intact.

	Quantum attack	Effect	Fix
Symmetric (AES, ChaCha20)	Grover	strength halved	bigger keys (AES-256)
Public-key (RSA, DH, ECDSA)	Shor	broken	new math (below)

11.2 Harvest now, decrypt later

The machine is years off, but for *confidentiality* the threat is already present. An adversary can record your encrypted traffic today and store it until a quantum computer exists, then go back and break the key exchange that protected it. So anything that must stay secret for a decade needs post-quantum key agreement *now*.

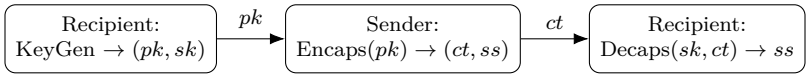
Authentication is less urgent. You cannot forge a signature on a past message after the fact, so post-quantum signatures only need to be in place by the time quantum computers actually arrive — not before.

11.3 The replacements

Post-quantum cryptography rebuilds key agreement and signatures on problems believed hard even for quantum computers — mostly *lattice* problems (finding short vectors in high-dimensional lattices, e.g. Learning With Errors), for which no efficient quantum algorithm is known. NIST standardised three in 2024:

Classical (broken)	Post-quantum replacement	Based on
(EC)DH key agreement (§8)	ML-KEM / Kyber (FIPS 203)	lattices (LWE)
RSA, ECDSA signatures (§9)	ML-DSA / Dilithium (FIPS 204)	lattices
(a conservative backup)	SLH-DSA / SPHINCS$^+$ (FIPS 205)	hash functions only

The new key agreement is shaped differently from Diffie–Hellman: it is a *KEM* (key encapsulation mechanism), not a mutual exchange of shares. The recipient publishes a public key; the sender *encapsulates* a random shared secret to it, producing a ciphertext; the recipient *decapsulates* that ciphertext with the private key to recover the same secret.

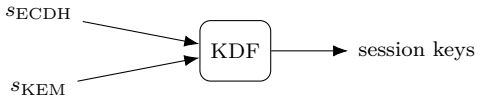


Both sides end up with the shared secret ss, which feeds a KDF exactly as Diffie–Hellman's s did (§8) — same outcome, different machinery. (These schemes lean on SHAKE/Keccak internally — the same sponge from the hashing collection.)

11.4 Hybrid, for now

Deployments today do not yet bet the channel on the young lattice schemes alone. They go *hybrid*: run a classical ECDH (§8) *and* a post-quantum KEM together, and feed *both* shared secrets into the KDF:

$$k = \mathrm{KDF}\big(s_{\mathrm{ECDH}} \,\|\, s_{\mathrm{KEM}}\big).$$



You are safe if *either* holds: the post-quantum part defeats harvest-now-decrypt -later, while the battle-tested ECDH part guards against a flaw later found in the new scheme.

11.5 Symbols

No new persistent symbols. Local to this section:

Local symbol	Meaning (this section)
KEM	key encapsulation mechanism (post-quantum key agreement)
pk, sk	the recipient's public and private KEM keys
ct, ss	the encapsulated ciphertext and the shared secret it carries
s_{ECDH}, s_{KEM}	the classical and post-quantum shared secrets (hybrid)

12 Randomness

This document has quietly rested on two words, over and over: "pick at random." A random key (§1), a fresh random nonce (§2, §7), random Diffie–Hellman secrets a, b (§8), a fresh per-signature value (§9). All of that security assumes those values are *unpredictable*.

If an attacker can predict or even narrow down your "random" choice, they recover the key or secret directly — without touching AES, without factoring, without any of the hard problems. Bad randomness is a side door around all the strong math. So it is worth asking where the randomness actually comes from.

12.1 Random means unpredictable

Cryptographic randomness is not about *statistical* prettiness — passing dice-roll uniformity tests. It is about *unpredictability*: given everything an attacker knows, including all previous outputs, they must not guess the next bit better than chance.

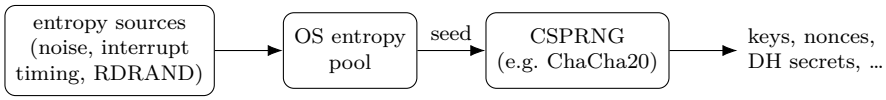
A sequence can be statistically flawless yet completely predictable — the digits of π, or the output of a simple linear congruential generator — and is then worthless for keys. "Looks random" is not "is random."

12.2 Entropy plus a CSPRNG

Real systems get there with two ingredients.

Entropy is genuine physical unpredictability the machine harvests: thermal and electrical noise, the jitter in the timing of interrupts (keystrokes, disk, network), or a dedicated hardware generator (Intel's RDRAND). It is truly unpredictable, but slow and scarce.

A *CSPRNG* (cryptographically secure pseudo-random number generator) is a deterministic algorithm that stretches a small high-entropy *seed* into an endless stream of unpredictable-looking bits, built from a crypto primitive. Predicting its output without the seed would mean breaking that primitive.



The operating system wires this up for you: it gathers entropy, seeds a CSPRNG, and serves the bits through a system call — `getrandom()` or `/dev/urandom` on Linux, `BCryptGenRandom` on Windows, `SecRandomCopyBytes` on macOS. Draw every key, nonce, and secret from there.

A CSPRNG is, in fact, essentially a stream cipher run for its keystream: the seed plays the role of key-plus-nonce, and the output is the keystream. ChaCha20 (§5) is exactly this shape — and Linux's `/dev/urandom` is built on ChaCha20. The same ARX engine that encrypts also manufactures randomness.

12.3 When randomness fails

Strong cipher, broken generator is one of the worst failure modes there is — and it keeps happening.

Debian OpenSSL (2008). A cleanup removed the line that mixed entropy into the seed, leaving essentially just the process ID — about 2^{15} possibilities. Every key Debian generated for roughly

two years was one of ∼32,000, all enumerable. The ciphers were fine; the randomness was not.

ECDSA nonces (Sony PS3, 2010*; some Bitcoin wallets).* This is §9's footgun in the wild: Sony used a *constant* value where a fresh random one was required, and weak wallet generators produced repeated or biased nonces. Either way the private key fell straight out — a randomness failure, not a flaw in the signature scheme.

Dual_EC_DRBG (∼2007*).* A NIST-standardised CSPRNG whose elliptic-curve constants could be chosen so that whoever knew a secret relationship could predict all of its output — a suspected backdoor. It is the mirror image of §5's nothing-up-my-sleeve constants: a generator is only as trustworthy as the provenance of its magic numbers.

12.4 In practice

Never roll your own, and never use a *non-cryptographic* PRNG — C's `rand()`, a Mersenne Twister, a linear congruential generator — for anything secret. They are built for speed and uniformity, not unpredictability, and are trivially predicted from a few outputs.

Use the operating system's CSPRNG (`getrandom`, `/dev/urandom`, `BCryptGenRandom`), or a vetted library that wraps it.

Watch the low-entropy moments: a device at first boot, a freshly imaged embedded board, or a just-cloned virtual machine may have gathered little entropy and hand out weak — or duplicated — randomness. A cloned VM that copies the CSPRNG state will reissue the very same "random" nonces, which is the clone hazard already flagged in §7.

12.5 Symbols

No new persistent symbols. Local to this section:

Local symbol	Meaning (this section)
entropy	genuine physical unpredictability the machine harvests
CSPRNG	cryptographically secure generator: seed → unpredictable stream
seed	the high-entropy value that initialises the CSPRNG

13 Public-key encryption

We have met two public-key tools: key agreement (Diffie–Hellman, §8) and signatures (§9). The third is *public-key encryption* — scrambling a message so that only the holder of a private key can read it.

13.1 The third public-key tool

Each recipient has a key pair: a *public key* pk anyone may use to encrypt, and a *private key* sk only they hold to decrypt.

$$c = \mathrm{Enc}_{pk}(m), \qquad m = \mathrm{Dec}_{sk}(c).$$

Anyone with pk can encrypt to the recipient; only sk can recover the message.

This is the mirror of signatures (§9). Signing uses the *private* key and verifying the *public* one; encryption is the other way around — the *public* key encrypts, the *private* key decrypts. (A party usually keeps *separate* pairs for signing and for encryption.)

Unlike Diffie–Hellman, it is one-shot and non-interactive: there is no handshake and the recipient need not be online. You encrypt to their published pk whenever you like.

13.2 Textbook RSA, and why it needs padding

RSA can encrypt directly, reusing the modulus $N = pq$ and exponents of §9 — but with the roles swapped from signing:

$$c = m^e \bmod N \quad (\text{public } e,\ \text{anyone}), \qquad m = c^d \bmod N \quad (\text{private } d).$$

The factoring trapdoor protects it, just as for signatures.

But raw "textbook" RSA is unsafe. It is deterministic — the same message always gives the same ciphertext, leaking equality, and a small or guessable message space can simply be tried — and it is malleable. Real RSA encryption first pads the message with randomness (*RSA-OAEP*), exactly as signing uses PSS. Never encrypt with bare RSA.

13.3 Why you don't encrypt the data directly

Two hard limits stop you from public-key-encrypting a whole message.

Public-key operations are *slow* — orders of magnitude slower than AES (§6) or ChaCha20 (§5).

And they are *size-limited*: RSA can only encrypt a message smaller than its modulus (RSA-2048 tops out around 256 bytes). You cannot wrap a large file this way at all.

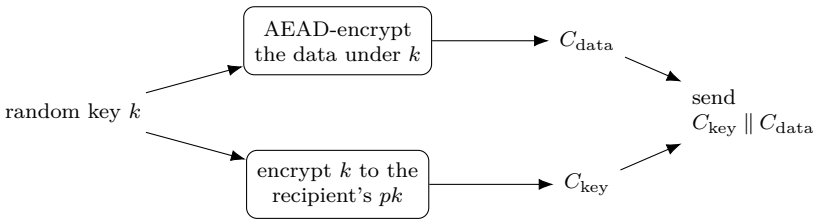
So public-key encryption is used to protect one small thing — a key — and symmetric crypto carries the bulk. That is *hybrid encryption*.

13.4 Hybrid encryption

The recipe:

1. generate a fresh random symmetric key k (a per-message "content key");
2. encrypt the data with k using a fast AEAD (§3) — call it C_{data};

3. encrypt k to the recipient's public key — call it C_{key};
4. send $C_{\text{key}} \,\|\, C_{\text{data}}$.



The recipient runs $\text{Dec}_{sk}(C_{\text{key}})$ to recover k, then AEAD-decrypts C_{data} with it.

The payoff is the best of both worlds: public-key *convenience* (encrypt to anyone's pk, with no prior shared secret and no handshake) with symmetric *speed* (the bulk data goes through AEAD). It is how PGP/GPG, age, S/MIME, and libsodium sealed boxes encrypt files and mail — and it is the data phase of TLS once the handshake of §10 has settled on a key.

13.5 The modern shape: KEM + DEM

Today the pattern is named in two halves:

- a **KEM** (key encapsulation mechanism) delivers the symmetric key — classic RSA-style "pick k and encrypt it" is one KEM, while the cleaner modern KEMs (including the post-quantum ML-KEM of §11) instead $\text{Encaps}(pk) \to (ct, ss)$ and let the recipient $\text{Decaps}(sk, ct) \to ss$, then derive the key with a KDF;
- a **DEM** (data encapsulation mechanism) is just the AEAD that seals the data under that key.

So hybrid encryption is exactly KEM + DEM — the same encapsulation idea as §11, applied here to encrypt to a public key rather than to run an interactive exchange.

13.6 Who sent it?

Encryption hides the message but proves nothing about its origin — *anyone* can encrypt to your public key. To authenticate the sender you add a signature (§9) over the message or ciphertext. (An anonymous "sealed box" gives no sender authentication; an authenticated box binds a specific sender.) Confidentiality and authenticity remain separate goals, exactly as in §3.

13.7 Symbols

No new persistent symbols. Local to this section:

Local symbol	Meaning (this section)
pk, sk	recipient's public / private keys (separate from signing)
Enc_{pk}, Dec_{sk}	public-key encrypt / decrypt
C_{key}, C_{data}	PK-encrypted content key; AEAD-encrypted data
KEM, DEM	key- / data-encapsulation mechanism (two halves of hybrid)
N, e, d	RSA modulus, public and private exponents (§9)

14 Inside elliptic curves (ECDH)

Diffie–Hellman (§8) and signatures (§9) work in *any* group where the discrete logarithm is hard. We have kept pointing at elliptic curves as "the same idea in a different group, with smaller keys." This section opens that group up.

14.1 Why elliptic curves

Classical Diffie–Hellman lives in the integers mod a prime and needs ∼3072-bit keys. An elliptic curve gives a group where the discrete log is far harder *per bit*, so a 256-bit key matches a 3072-bit classical one — smaller and faster. That is why curves (X25519 in §8, Ed25519 in §9) are the modern default.

14.2 Adding points: the group

An elliptic curve over a finite field is the set of points (x, y) — with x, y in $\mathbb{F}_p$ (integers mod a prime p) — satisfying

$$y^2 = x^3 + \alpha x + \beta,$$

together with one extra "point at infinity" $\mathcal{O}$.

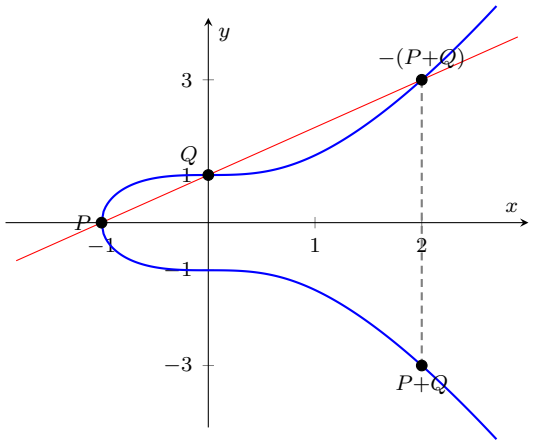
Because the equation contains y^2, whenever (x, y) lies on the curve so does $(x, -y)$ — the curve is symmetric about the x-axis. That is precisely why negation is a reflection across that axis, $-(x, y) = (x, -y)$. (This holds for the short Weierstrass form used over a prime field; the Edwards form of §18 is instead symmetric about the y-axis, with $-(x, y) = (-x, y)$.)

These points form a group under a geometric *chord-and-tangent* rule:

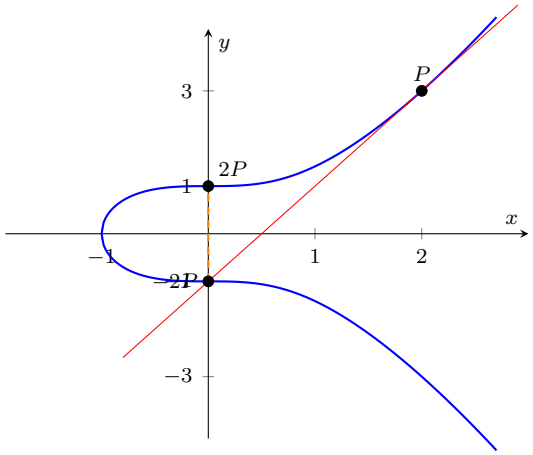
- **Add** $P + Q$ (distinct points): draw the straight line through P and Q; it meets the curve at exactly one more point; reflect that across the x-axis — the reflection is $P + Q$.
- **Double** $2P = P + P$: use the *tangent* at P instead of a chord; it meets the curve once more; reflect to get $2P$.
- $\mathcal{O}$ is the identity ($P + \mathcal{O} = P$), and the reflection of P across the x-axis is its inverse $-P$ (so $P + (-P) = \mathcal{O}$).

To make the rule concrete, take the curve $y^2 = x^3 + 1$ over the reals.

Addition. With $P = (-1, 0)$ and $Q = (0, 1)$, the line through them is $y = x + 1$; it meets the curve again at $(2, 3)$, and reflecting that across the x-axis gives $P + Q = (2, -3)$.



Doubling. With $P = (2, 3)$, the *tangent* there is $y = 2x - 1$; it meets the curve again at $(0, -1)$, and reflecting gives $2P = (0, 1)$.



The result is an abelian group — exactly the structure §8 and §9 need. (Over $\mathbb{F}_p$ the "geometry" is really the same operations done with modular arithmetic, but the chord-and-tangent rule is the picture to keep in mind.)

14.3 Scalar multiplication: the one-way step

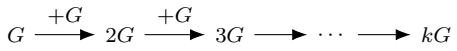
Adding P to itself k times gives *scalar multiplication*

$$kP = \underbrace{P + P + \cdots + P}_{k \text{ times}},$$

the curve's analogue of the exponentiation g^k of §8.

Each $+$ in that sum is the chord-and-tangent addition defined above: $2P$ is a tangent (doubling) step, $3P = 2P + P$ a chord step, and so on. So scalar multiplication is nothing but that one geometric operation, repeated.

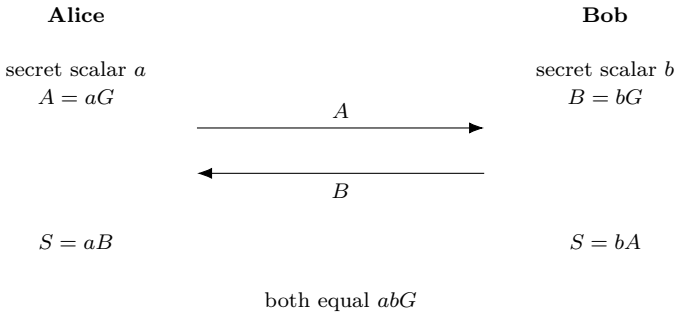
Over a finite field the multiples of a point eventually cycle back: the smallest n with $nG = \mathcal{O}$ is the *order* of G, and its n distinct multiples $G, 2G, \ldots, (n-1)G, \mathcal{O}$ form the cyclic group we compute in. Scalars are therefore taken mod n; a *prime* order n is preferred, as it leaves no smaller subgroups for an attacker to exploit.



Forward is cheap: *double-and-add* (the additive twin of square-and-multiply) computes kP in about $\log k$ steps, so kP is fast even for an enormous k. Backward is hard: given P and kP, recovering k is the *elliptic-curve discrete logarithm problem* (ECDLP). The best classical attacks cost about $\sqrt{\text{group size}}$, so a 256-bit curve gives ∼128-bit security — the reason the keys are so small.

14.4 ECDH

Elliptic-curve Diffie–Hellman is just §8's protocol run in this group. Public to all: a curve and a base point G.



Both land on the same point because the scalars commute: $aB = a(bG) = abG = b(aG) = bA$. An eavesdropper sees G, aG, bG, but reaching abG needs a or b — the ECDLP. The shared secret is a point; you take its x-coordinate and run it through a KDF to get the key, exactly as $s \to k$ in §8.

It is structurally identical to classical DH; only the group changed:

	Classical DH (§8)	ECDH
Group	integers mod p	points on a curve
Mixing	$g^a \bmod p$	aG (scalar mult)
Secret	exponent a	scalar a
Public	g^a	aG
Shared	g^{ab}	abG
Hard problem	discrete log	ECDLP
Key size (∼128-bit)	∼3072-bit	∼256-bit

Notation flag. Here p is the field prime (as in §8), and P, Q, R denote curve *points* — not the plaintext P of earlier sections; the curve's constants are α, β.

14.5 Curve25519 / X25519 in practice

The dominant instantiation is *Curve25519*, a specific curve over the prime $p = 2^{255} - 19$, used by the *X25519* function (ECDH on that curve).

It was chosen for speed and for safety: its parameters are rigid and transparently derived (nothing-up-my-sleeve, like the constants of §5 and §12), and it avoids several common implementation pitfalls.

X25519 is the modern default — TLS, SSH, Signal, WireGuard — and its Edwards-curve cousin powers the Ed25519 signatures of §9.

14.6 Symbols

No new persistent symbols. Local to this section:

Local symbol	Meaning (this section)
$\mathbb{F}_p$	the finite field of integers mod a prime p
$\alpha,\ \beta$	the curve's constants in $y^2 = x^3 + \alpha x + \beta$
P, Q, R	curve *points* (not the plaintext of earlier sections)
$\mathcal{O}$	the point at infinity — the group identity
G	the public base point (generator)
a, b	Alice's and Bob's secret scalars
kP	scalar multiplication: P added to itself k times
ECDLP	elliptic-curve discrete logarithm problem (the hard one)

15 Side-channel attacks

§12 showed one way sound cryptography fails in practice: bad randomness. Side channels are the other. The cipher is mathematically unbroken — but its *execution* on a real machine leaks.

15.1 Attacking the implementation, not the algorithm

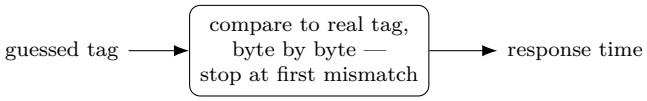
An attacker need not touch the front door — the inputs and outputs the math protects. Instead they watch a *side effect* of the computation: how long it took, what it did to the cache, how much power it drew, what it radiated. If any of those depends on secret data, the secret leaks.

These attacks are dangerous precisely because they sidestep the hard problems entirely. No factoring, no discrete log — just measurement.

15.2 Timing attacks

If an operation's *duration* depends on a secret, a stopwatch leaks the secret.

The canonical case is the one §3 warned about: comparing a received tag to the real one with an ordinary byte-by-byte compare that *returns at the first mismatch*. Then the response time grows with the number of matching leading bytes.



longer time ⇒ one more leading byte matched
⇒ keep that byte, then attack the next

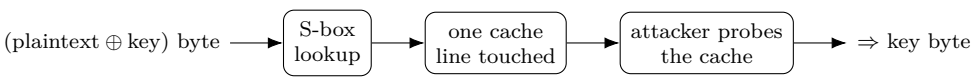
So an attacker forges a tag byte by byte: try all 256 values for byte 1, keep the one that runs slightly longer, move to byte 2, and so on — turning an infeasible 2^{128} guess into a few thousand tries.

The same trap hides in any secret-dependent branch (an `if` on a key bit), and in variable-time arithmetic — notably a square-and-multiply (§8) or double-and-add (§14) that skips steps on zero bits, leaking the private exponent or scalar. These work even remotely: RSA keys have been extracted over a network from timing alone.

15.3 Cache-timing attacks

A subtler timing channel runs through the CPU cache: data already cached is fast to read, data that must be fetched is slow. So if a computation reads a table at a *secret-dependent index*, whether that entry is cached leaks which index was used.

The classic target is AES's S-box (§6), implemented as a 256-byte lookup indexed by (plaintext ⊕ key) bytes:



An attacker sharing the machine primes the cache, lets AES run, then times their own reloads (the Flush+Reload technique) to see which table lines AES touched — recovering key bytes. This is exactly the weakness §5 and §6 named: it is why ChaCha20's table-free ARX is preferred in

software, and why AES is safest on the AES-NI hardware instructions, which run a round with no data-dependent lookup.

15.4 Power and electromagnetic analysis

For a device the attacker can hold — a smart card, a hardware wallet, an embedded chip — the power it draws and the electromagnetic field it emits both track the operations and data inside.

Simple power analysis (SPA) reads the secret straight off one trace: a square-and-multiply leaves a visible pattern whose shape spells out the exponent bits.

Differential power analysis (DPA) is stronger — collect many traces and statistically correlate them against guesses of an intermediate value; the correct key guess stands out even when buried in noise.

(Related are *fault attacks*: deliberately glitch the chip — a voltage or clock spike, a laser — to force a wrong result that leaks the key. A single faulty RSA signature can reveal the factorisation.)

15.5 The defence: constant-time

One principle ties the software defences together: a crypto operation's running time and memory-access pattern must not depend on secret data — *constant-time* (more precisely, data-independent) code.

In practice that means: no secret-dependent branches (compute both sides and select with arithmetic); no secret-dependent memory addresses (no `table[secret]` — use ARX, bit-slicing, or hardware AES); and the constant-time comparison §3 required (XOR all the bytes together, test once at the end).

Design choices carry much of the load: ARX ciphers (ChaCha20, §5) have no tables to leak, X25519 (§14) was built for safe constant-time code, and AES-NI removes the S-box table. Power and EM, by contrast, need hardware countermeasures — masking (split each secret into random shares so no single measured point depends on it), plus shielding and added noise.

Channel	What it leaks	Defence
Timing	duration depends on the secret	no secret branches; constant-time compare
Cache	which table index was used	no secret-indexed tables (ARX, AES-NI)
Power / EM	the data and ops a chip runs	masking, shielding

The lesson is that security is a property of the whole system. "The cipher is unbroken" does not mean "your implementation is safe" — which is why constant-time has shadowed this entire document.

15.6 Symbols

No new persistent symbols. Local to this section:

Local term	Meaning (this section)
side channel	an observable effect (time, cache, power, EM) that depends on secrets
constant-time	code whose timing and memory access do not depend on secret data
SPA, DPA	simple / differential power analysis
Flush+Reload	a cache-probing technique behind cache-timing attacks

16 Key management

Every section so far opened with "a key" and assumed it was secret and present. But a key is an object with a *lifecycle* — it must be generated, stored, used, rotated, and destroyed — and in the real world most failures live here, not in the algorithms.

16.1 The forgotten layer

Keys committed to source control, left in logs, baked into firmware, never rotated, or lifted from an unprotected device — these break far more systems than any cipher weakness. The strongest algorithm in this document is worthless the moment its key leaks. This section is the operational layer beneath the math.

16.2 Generating keys

A symmetric key is just random bytes, drawn from a CSPRNG (§12) at the algorithm's length (e.g. 256 bits for AES-256). Never from a weak generator, and never use a password *directly* as a key.

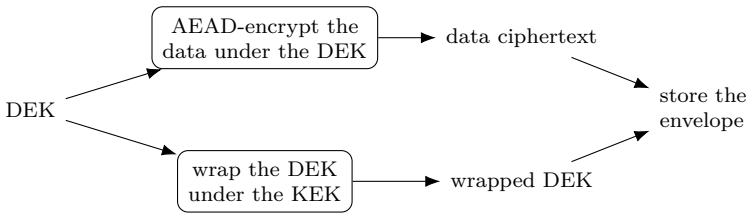
A password must first pass through a slow, salted password KDF (Argon2 or PBKDF2, from the hashing collection), which stretches a low-entropy secret into a key while resisting guessing.

Asymmetric keys are generated as a pair (§9, §13, §14): the private key from the CSPRNG, the public key derived from it.

16.3 Storage: the key hierarchy

Where should a key live so it is not simply sitting in a readable file? On a spectrum: in memory only (ephemeral session keys, §10 — safest, but gone on restart); encrypted on disk under another key; in a secrets manager; or in *hardware* — an HSM, TPM, or secure enclave, where the key never leaves. You feed data in and get results out, so even a compromised host (or a side channel, §15) cannot extract it.

But "encrypt the key under another key" only moves the problem — eventually some root key must rest in hardware or a human secret. The standard way to organise this is a key hierarchy with *envelope encryption*: a long-lived *key-encryption key* (KEK) wraps many short-lived *data-encryption keys* (DEKs); each DEK encrypts the actual data; and you store each data ciphertext alongside its wrapped DEK — the "envelope."



KEK: kept in an HSM / KMS, never exported

To read the data, unwrap the DEK with the KEK, then decrypt. Only the small KEK needs the strongest protection (an HSM or cloud KMS that never exports it); DEKs can be plentiful, per-file, generated freely, and rotated cheaply. This is exactly how cloud KMS services operate.

16.4 Rotation and destruction

Keys should be replaced periodically, and immediately on suspected compromise, for three reasons: to bound how much data rides on any one key (cryptographic wear — recall GCM's per-key message limit, §7), to limit the blast radius if a key leaks, and to meet policy.

Envelope encryption makes rotation cheap: to rotate the KEK, just re-wrap the DEKs under the new one — the bulk data, still under unchanged DEKs, is never touched. Tag each ciphertext with a key identifier so you know which key opens it.

To retire a key for good, destroy every copy. *Crypto-shredding* turns this into a feature: delete the key and the data it protected becomes permanently unreadable — a way to "erase" terabytes by erasing a few bytes.

For public keys, retire a compromised one by revoking its certificate (§9) — through revocation lists or OCSP — so it is no longer trusted.

16.5 Principles

A few rules underlie all of the above:

- **One key, one job.** Separate keys per purpose — encryption vs. MAC ($k_e \neq k_m$, §3), per-direction session keys (§10). A key's exposure must not spill across roles.
- **Minimise exposure.** Keys live in as few places as possible, ideally hardware; never written to logs, never committed to source control.
- **Plan for compromise.** Assume a key will eventually leak, and keep rotation, revocation, and crypto-shredding ready so the damage stays bounded.

16.6 Symbols

No new persistent symbols. Local to this section:

Local term	Meaning (this section)
KEK	key-encryption key: long-lived, wraps DEKs, kept in hardware
DEK	data-encryption key: per-data, encrypts the actual content
wrap / unwrap	encrypt / decrypt one key under another
HSM, KMS	hardware security module / key-management service
crypto-shredding	destroying data by destroying its key

17 The padding-oracle attack

§3 claimed that decrypting attacker-supplied ciphertext *before* verifying it is dangerous, and that MAC-then-Encrypt opened the door to "padding-oracle attacks." Here is that attack in full — decrypting CBC ciphertext byte by byte, with *no key*, from a single bit of feedback.

17.1 The setup: CBC, padding, and an oracle

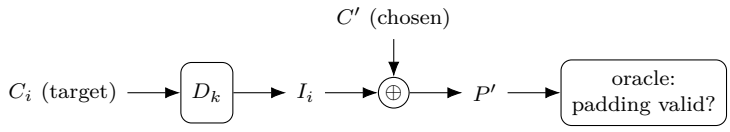
Recall CBC decryption (§2): $P_i = D_k(C_i) \oplus C_{i-1}$, with $C_0 = IV$. And PKCS#7 padding (§2.4): the last block ends with p bytes each equal to p, and on decryption the receiver checks that and rejects bad padding.

A *padding oracle* is anything that tells the attacker whether a submitted ciphertext decrypted to *valid padding* — a distinct error, a different response, or even a timing difference (a side channel, §15). That is one bit per query. It is enough.

17.2 The key idea: control the previous block

Write $I_i = D_k(C_i)$ for the *intermediate value* — the block cipher's decryption of C_i, *before* the XOR — so that $P_i = I_i \oplus C_{i-1}$. Crucially, I_i depends only on C_i and the key, so it is *fixed* for a given target block, even though the attacker cannot compute it.

The attacker submits a two-block ciphertext: a *chosen* block C' followed by the real target C_i. The decryptor produces, as the last "plaintext" block, $P' = I_i \oplus C'$ — and checks *its* padding.



Because the attacker controls C' byte by byte, they steer $P' = I_i \oplus C'$ (offset by the unknown I_i). So if they can learn I_i, the real plaintext follows for free: $P_i = I_i \oplus C_{i-1}$, and C_{i-1} is just known ciphertext.

17.3 Cracking the last byte

Vary the last byte of C' through all 256 values. For exactly one, the last byte of P' becomes `0x01` — valid one-byte padding — and the oracle says "valid." Then

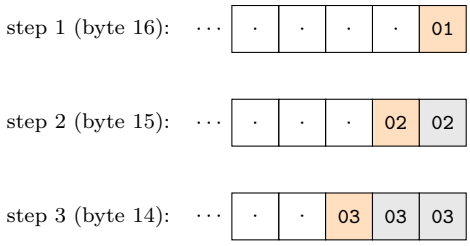
$$P'[16] = I_i[16] \oplus C'[16] = \texttt{0x01} \implies I_i[16] = C'[16] \oplus \texttt{0x01}.$$

One byte of the intermediate value, recovered in at most 256 tries.

(Caveat: very rarely the "valid" answer is really a longer padding such as `0x02 0x02`, a false positive. Confirm by also flipping the second-to-last byte: genuine `0x01` padding is unaffected, a coincidental longer one breaks. A couple of extra queries settle it.)

17.4 Cracking the rest

Now climb inward. To get the next byte, aim for two-byte padding `0x02 0x02`: using the known $I_i[16]$, set $C'[16] = I_i[16] \oplus \texttt{0x02}$ so $P'[16] = \texttt{0x02}$, then brute-force $C'[15]$ until the oracle accepts. In general, to recover the k-th byte from the end, force the already-known trailing bytes to the padding value k and search the next byte.



Orange is the byte being brute-forced to the shown value; gray is bytes already recovered, forced to it. Sixteen bytes fall in at most $256 \times 16 \approx 4096$ queries; then $P_i = I_i \oplus C_{i-1}$ gives the real block, and repeating per block decrypts the whole message — never once knowing the key.

17.5 Why it works, and the fix

The decryptor became an oracle: on attacker-supplied ciphertext it leaked a function of the secret intermediate value (was the padding valid?). That is exactly the failure §3 warned of — inspecting attacker data *after* decrypting but *before* (or instead of) verifying its integrity, the weakness of MAC-then-Encrypt.

The fix is authenticated encryption / Encrypt-then-MAC (§3): verify the tag over the ciphertext *first* and reject forgeries with $\perp$ before any decryption or padding check. A forged ciphertext never reaches the padding logic, so there is no oracle.

Merely hiding the signal — uniform error messages, constant-time padding checks (§15) — is only a band-aid: the Lucky 13 attack recovered the bit from timing alone. The real cure is AEAD.

This is not academic. Vaudenay introduced the attack in 2002; padding oracles went on to break SSL/TLS in CBC mode (POODLE, Lucky 13), ASP.NET, and more. It is the canonical reason modern systems use AEAD and never unauthenticated CBC.

17.6 Symbols

No new persistent symbols. Local to this section:

Local symbol	Meaning (this section)
$C_i,\ C_{i-1}$	the target ciphertext block and the one before it (§2)
I_i	the intermediate value $D_k(C_i)$, before the CBC XOR
C'	the attacker's chosen "previous" block
P'	the last plaintext block the oracle checks, $P' = I_i \oplus C'$
padding oracle	a signal revealing whether decryption gave valid padding
k	the padding length being forged at each step

18 The 25519 curves: X25519 and Ed25519

§14 explained elliptic curves in general. In practice you do not invent a curve — you use a vetted, named one, and today that is overwhelmingly the *25519* family: X25519 for key agreement and Ed25519 for signatures.

18.1 Why a named curve

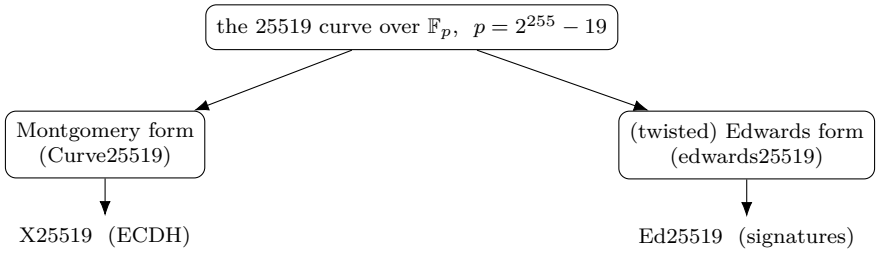
Rolling your own curve is as dangerous as rolling your own cipher. A bad choice can hide a weakness, and even a good curve is easy to implement insecurely. So everyone shares a few standard curves whose parameters and pitfalls are well studied.

The 25519 family (Bernstein) is the modern default for three reasons: it is fast, its parameters are *rigid* — derived transparently as the smallest values meeting the security criteria, a nothing-up-my-sleeve choice (§12) — and it is built to be hard to implement wrong. That contrasts with the older NIST P-curves, whose coefficients come from an unexplained seed, a provenance that bred distrust after Dual_EC (§12).

18.2 One curve, two forms

The name comes from the field: all arithmetic is mod the prime $p = 2^{255} - 19$ (a prime just below a power of two, which makes reduction fast), giving about 128-bit security with 32-byte keys.

The same underlying curve is written in two equivalent shapes, each suited to one job:



The *Montgomery* form (Curve25519) admits a fast, constant-time, x-coordinate-only scalar multiplication (the "Montgomery ladder"), ideal for ECDH. The *Edwards* form (edwards25519) has a *complete* addition law — one formula that works for every pair of points, with no special cases to leak timing — ideal for signatures. The two forms are the same curve under a change of coordinates.

18.3 X25519: key agreement

X25519 is a function, not a full point API. It takes a scalar and a u-coordinate (the Montgomery x) and returns a u-coordinate; public keys and shared secrets are just 32-byte u-values. It is ECDH (§14) with a deliberately small, safe interface.

With base point $u = 9$, Alice (scalar d_A) and Bob (scalar d_B) do:

$$\text{pub}_A = \text{X25519}(d_A, 9), \qquad \text{shared} = \text{X25519}(d_A, \text{pub}_B) = \text{X25519}(d_B, \text{pub}_A).$$

Two safety features are built in. The scalar is *clamped* — a few bits forced — so it is a multiple of the cofactor and in a fixed range, sidestepping small-subgroup attacks. And every 32-byte string is an acceptable public key, so there are no invalid-curve checks to forget. This misuse-resistance is why X25519 is the default key exchange in TLS 1.3, SSH, Signal, and WireGuard.

18.4 ECDSA: the scheme Ed25519 improves on

To see why Ed25519 is built the way it is, look first at the older standard it replaces — ECDSA, the Elliptic Curve Digital Signature Algorithm. It uses the elliptic-curve key of §14: the private key is a secret integer (a *scalar*) d, and the public key is the *point* $Q = dG$ — the base point G added to itself d times (scalar multiplication, §14), easy to compute but infeasible to reverse. Here n is the order of G (§14) — the length of the cycle of multiples of G — so scalars are taken mod n. To sign a message M, hash it to a number $e = H(M)$ and pick a *fresh random* secret nonce k ($1 \le k < n$):

$$R = kG, \quad r = (x\text{-coordinate of } R) \bmod n, \quad s = k^{-1} \cdot (e + r\,d) \bmod n.$$

All of this is *modular* arithmetic — integers in $\{0, 1, \ldots, n-1\}$ taken mod n, never floating point. Juxtaposition like $r\,d$ is ordinary multiplication $r \cdot d$. And k^{-1} is the *modular inverse* of k *alone* — the unique integer with $k \cdot k^{-1} \equiv 1 \pmod{n}$ (found by the extended Euclidean algorithm, not a fraction) — which is *then multiplied* by the number $(e + r\,d)$. It is not the inverse of the whole product.

So computing s is two steps: first find k^{-1}, then form $k^{-1} \cdot (e + r\,d) \bmod n$. For instance with $n = 7$, $k = 3$, $e = 2$, $r = 4$, $d = 5$: the inverse is $k^{-1} = 5$ (since $3 \cdot 5 = 15 \equiv 1$), and $e + r\,d = 2 + 20 = 22$, so $s = 5 \cdot 22 = 110 \equiv 5 \pmod 7$. Because n is prime, every nonzero value has an inverse, so s is always an integer in $\{0, \ldots, n-1\}$.

The signature is the pair (r, s). To verify with Q, compute $u_1 = e\,s^{-1}$ and $u_2 = r\,s^{-1} \pmod n$, and accept iff the x-coordinate of $u_1G + u_2Q$, reduced mod n, equals r — which works because $u_1G + u_2Q = s^{-1}(e + rd)\,G = kG = R$.

The whole danger lives in the nonce k: it must be unpredictable and used once. Suppose two signatures reuse the same k (so they share the same r):

$$s_1 = k^{-1}(e_1 + rd), \qquad s_2 = k^{-1}(e_2 + rd).$$

Subtracting clears d: $s_1 - s_2 = k^{-1}(e_1 - e_2)$, so $k = (e_1 - e_2)/(s_1 - s_2) \bmod n$ — the attacker reads the nonce straight off two public signatures. And once k is known, $d = (s_1 k - e_1)/r \bmod n$ hands over the *private key*. Even a slightly biased or partly-leaked k recovers d by lattice methods. This is the footgun of §9 and §12, now in full: the nonce sits in an equation *with* the private key, so any weakness in k becomes a weakness in d.

18.5 EdDSA and Ed25519: deterministic signatures

EdDSA, the Edwards-curve Digital Signature Algorithm, is a different design — a *Schnorr* signature on a (twisted) Edwards curve — that closes ECDSA's two weak spots: signing needs no modular inverse, and the per-signature value is derived *deterministically* instead of chosen at random. Ed25519 is EdDSA on edwards25519 with SHA-512 as the hash H.

A private key is a 32-byte seed; hashing it splits into a clamped private scalar d and a secret *prefix*. The public key is the point $A = dB$, where B is the base point. To sign a message M:

$$r = H(\text{prefix} \,\|\, M), \quad R = rB, \quad h = H(R \,\|\, A \,\|\, M), \quad S = (r + h\,d) \bmod L,$$

with L the prime order of the base point. The signature is the pair (R, S) — 64 bytes. Verification accepts iff

$$S\,B = R + h\,A \qquad (\text{recomputing } h = H(R \,\|\, A \,\|\, M)),$$

since $SB = (r + hd)B = R + hA$. This shape — a commitment R, a challenge h, a response $S = r + hd$ — is exactly a Schnorr signature.

The decisive choice is that the per-signature value r is computed *deterministically* from the secret prefix and the message — there is *no random number generator at signing time*. That removes the ECDSA footgun at the root: with no random nonce to repeat or bias, the private-key-recovery disaster simply cannot occur.

	per-signature value	risk
ECDSA	a fresh random nonce k	reuse or bias $\rightarrow$ key recovery
Ed25519	deterministic $r = H(\text{prefix} \parallel M)$	none — no RNG involved

Together with small keys, fast constant-time arithmetic (the complete Edwards addition), and rigid parameters, this is why Ed25519 is the modern default signature in SSH, TLS 1.3, Signal, and software signing.

Notation flag. ECDSA above uses lowercase (r, s), random nonce k, public key Q, order n; Ed25519 uses uppercase (R, S), deterministic nonce r, challenge h, public key $A = dB$, base B, order L — all local, distinct from the A, B, d, s of earlier sections.

18.6 Symbols

No new persistent symbols. Local to this section:

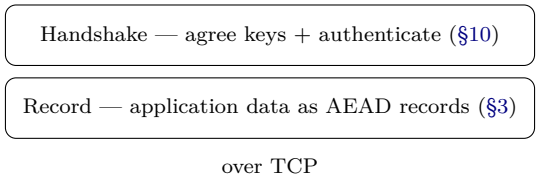
Local symbol	Meaning (this section)
p	the field prime $2^{255} - 19$
X25519	the ECDH function on Curve25519 (u-coordinates)
clamping	forcing a few scalar bits for safety (cofactor, range)
ECDSA	EC signature with a *random* nonce (§9)
$Q = dG,\ n$	ECDSA public key, and base-point order n
$e,\ k,\ (r, s)$	ECDSA message hash, random nonce, signature
Ed25519	EdDSA signature on edwards25519 with SHA-512
$B,\ A,\ d$	Ed25519 base point, public key $A = dB$, private scalar d
$r,\ R$	Ed25519 deterministic nonce $r = H(\text{prefix} \parallel M)$, $R = rB$
$h,\ (R, S)$	challenge $h = H(R \parallel A \parallel M)$; signature $S = (r + hd) \bmod L$
L	the prime order of the base point

19 TLS and the web PKI

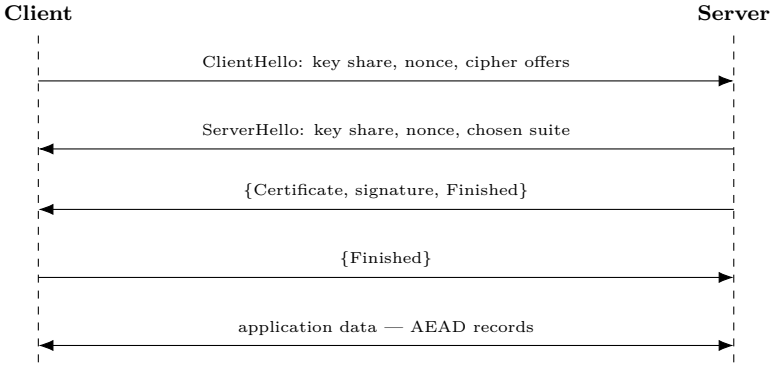
§10 built a secure channel (TLS its everyday example); §9 gave certificates. Here are the two real-world pieces: the TLS protocol *around* the handshake, and the *public-key infrastructure* (PKI) that decides whose certificates to trust.

19.1 TLS: the protocol around the handshake

Two layers — a *handshake* (agree keys, authenticate) and a *record* protocol (carry data as AEAD records):



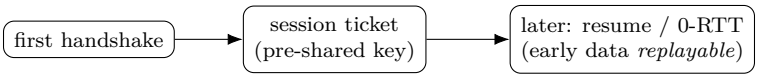
The TLS 1.3 handshake is one round trip; the client *offers* cipher suites and the server *picks* one ({ } marks encrypted messages):



The Finished MAC (§10) covers the whole transcript, so tampering with the negotiation — a *downgrade attack* — is caught.

	TLS 1.2	TLS 1.3
Round trips	2	1
Key exchange	RSA or (EC)DHE	(EC)DHE only
Old/weak options	CBC, RC4, RSA	removed
Handshake	mostly in clear	mostly encrypted

A returning client can *resume* from a ticket, or send *0-RTT* data in the first message — fast, but that early data is *replayable*, so it must be safe to repeat:



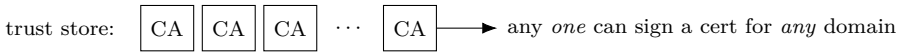
19.2 The web PKI: who to trust

A signature proves *who* signed; PKI says *whose* signature to trust. Your OS or browser ships a *trust store* of a few hundred root CAs; roots sign intermediates, intermediates sign the leaf (server) certificate — so the server presents a chain:

The browser verifies each link up to a trusted root. Certificates use the *X.509* format (public key + domain + validity + issuer signature). A domain-validated certificate proves only *control of the domain* — you place a CA-named token in DNS or at a URL (automated by ACME / Let's Encrypt) — not who you are.

19.3 The weak link, and the fixes

Any of the hundreds of trusted CAs can issue a certificate for *any* domain, so one breached CA can impersonate any site (DigiNotar, 2011: forged Google certificates, then removed from every trust store):



The fixes:

Fix	What it does
Short-lived certs	expire before a leak matters
Certificate Transparency	public append-only logs ⇒ mis-issuance is visible
CAA records	a DNS record limits which CAs may issue for a domain
Revocation (CRL / OCSP)	cancel a bad cert (in practice unreliable)

19.4 Symbols

No new persistent symbols. Local to this section:

Local term	Meaning (this section)
CA	certificate authority: an entity trusted to sign certificates
root / intermediate / leaf	the three levels of a certificate chain
trust store	the set of root CAs your OS or browser trusts by default
X.509	the certificate format binding a public key to an identity
CRL, OCSP	revocation: a published list, or an online status check
CT	Certificate Transparency: public append-only logs of issued certs
CAA	a DNS record naming which CAs may issue for a domain